
Why Cross-Skeleton Retargeting Is Non-Identifiable: Structural Limits of Generative Motion Models

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Abstract

Cross-skeleton motion generation trains generative models to carry action structure and motion intention from one body to another. Yet a target motion that shows the right action has two explanations that the training data cannot tell apart: the model transferred the source clip, or it recovered a typical motion for the requested action. We show that this ambiguity is structural rather than incidental: under standard generative objectives, the source-conditioned retargeting map is non-identifiable in sparse heterogeneous motion domains. Unpaired distribution matching yields *gauge non-identifiability*: the latent spaces of different skeletons can be transformed relative to one another without changing the training evidence, so different source-conditioned maps fit it equally well. Sparse paired supervision admits the complementary failure mode, *conditional-mean degeneration*: when clips are paired only by action, squared-error training converges to an average target motion that ignores the source clip. To make the missing evidence observable, we introduce *Source-Instance Fidelity* (SIF), a diagnostic that tests whether outputs differ from one another the way their source clips do, with the target skeleton and action held fixed. Under this diagnostic, methods that succeed at the standard action-level test on animal motion data often sit at the *source-blind floor*, while the methods that rise above it retain only a partial relational signal. Retargeting therefore needs objectives and evaluations that can identify the source-conditioned map it claims to learn. Project page: <https://keen-semolina-09811d.netlify.app/>.

1 Introduction

Motion generation increasingly has to work across bodies, not only on one fixed skeleton. Methods have moved from retargeting models that encode skeletal structure to generative and correspondence-based models built for many different bodies [Aberman et al., 2020, Lee et al., 2023, Wang et al., 2024, Gat et al., 2025, Chen et al., 2025b]. A motion clip may come from a human, a quadruped, or another animal, and the target body may differ in proportions, joint layout, and range of motion. Cross-skeleton retargeting asks for a motion on the target body that keeps both the action and the way the source clip performs it: its timing, its style, and the details that set it apart from other clips of the same action. We call this the motion intention of the source clip.

Progress in this area is usually measured by a retrieval test introduced with large multi-body motion libraries [Gat et al., 2025, Truebones Motions, 2025]. The generated motion serves as a query against real clips of the target body: clips with the requested action count as matches, and clips with other actions as non-matches. The score is the area under the receiver-operating-characteristic curve (AUC), the probability that a match ranks above a non-match. This is a sound test of whether the requested action can be recognized on the target body. It does not test whether the output came from the particular source clip that was given.

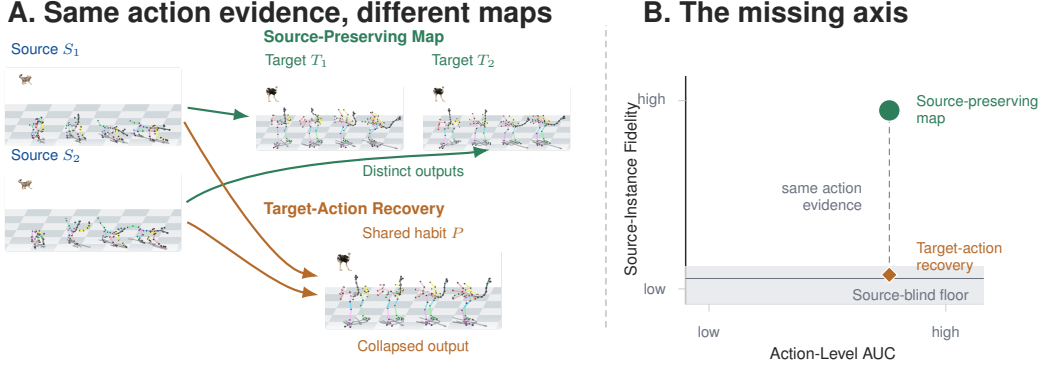


Figure 1: Action-level success does not identify the source-conditioned map. Two source clips with the same action can lead either to distinct target motions under a source-preserving map or to a shared target-action habit under source-blind recovery. Standard action-level AUC accepts both explanations; Source-Instance Fidelity tests whether the source-instance relation survives the change of body.

Retargeting makes a stronger claim: each source clip is carried to its own target motion. We call this mapping the source-conditioned map. Figure 1 shows why the retrieval score cannot check it. Two different coyote attacks should become two ostrich attacks that differ in the same way. A model that ignores the source and returns the same habitual ostrich attack for both, a target-action habit, produces outputs that are just as recognizable as attacks. The action-level score accepts both, and only a second question separates them: whether changing the source clip changes the output accordingly.

The ambiguity becomes structural in sparse libraries of many different bodies. In the animal motion data we study, the Truebones zoo [Truebones Motions, 2025], the skeletons differ in structure and joint count, most combinations of body and action have no clip at all, and many of the rest have only one. Such data show which actions each body can perform, but they contain no pairs that say how a given source clip should move on another body. The training data therefore do not determine which source-conditioned map is correct.

We show that this is a limit of identifiability: the training data are consistent with many different maps, and standard objectives give no reason to prefer one [Locatello et al., 2019, Khemakhem et al., 2020]. In this sense the source-conditioned map is *non-identifiable*. Two mechanisms produce the limit. When each body’s motions are matched only in distribution, through a latent space shared by all bodies, the latent space of one body can be rotated or otherwise transformed relative to another without changing the training loss, so different source-to-target maps fit the data equally well. We call this *gauge non-identifiability*. When training pairs each source clip with a randomly chosen target clip of the same action, squared-error training converges to the average target motion for that action, whatever the source. We call this *conditional-mean degeneration*. Both mechanisms explain why better action recovery does not by itself identify the retargeting map.

To measure what the retrieval score misses, we introduce *Source-Instance Fidelity* (SIF). For a fixed source body, target body, and action, SIF asks whether the outputs differ from one another the way their source clips do: two source clips that are far apart should lead to outputs that are far apart too. A model that ignores the source scores near zero, which we call the *source-blind floor*. SIF gives the true mapping a score near one on synthetic data with known correspondences, and a similarly high score on real human-to-robot pairs.

Measured this way, most methods fall short. On 1,891 combinations of source body, target body, and action from Truebones, nine of fourteen evaluated methods sit at or near the source-blind floor, including methods that score well on the retrieval test. The same reading also qualifies the methods that rise above the floor. ACE [Li et al., 2023], an adversarial cross-embodiment model that adds a motion-space constraint beyond per-skeleton distribution matching, retains only a partial relational signal: its outputs keep some of the ordering among source clips but vary little with them.

Retargeting therefore cannot be judged only by whether the target action is recognizable. A generated motion can look plausible and show the right action while ignoring how the source clip moved. What

has to be identified is the source-conditioned map itself, and the central question is which evidence and inductive biases are enough to select it.

2 Related Work

Cross-skeleton motion retargeting. The retargeting literature has progressively weakened the assumption that source and target bodies share a convenient representation. Early neural retargeting made skeletal structure explicit, while later skeleton-agnostic representations and context-conditioned sequence models sought motion spaces that remain meaningful across body graphs [Aberman et al., 2020, Lee et al., 2023, Wang et al., 2024]. More recent work carries the same ambition into sparse heterogeneous libraries, where generative models and sparse-correspondence procedures must operate without dense paired examples [Gat et al., 2025, Kim et al., 2025, Li et al., 2023, Chen et al., 2025b]. Our concern is orthogonal to this modeling trajectory: once an output has the requested action, the remaining question is whether the evidence identifies the particular source-conditioned map that retargeting claims.

Motion generation beyond fixed bodies. Recent motion generation work also shows that the fixed human skeleton is no longer the only natural domain of synthesis. The shared pressure across this broader literature is to represent motion when body identity, morphology, and topology become part of the conditioning problem rather than fixed background structure [Raab et al., 2024, Lee et al., 2025, Egan et al., 2024, Chen et al., 2025a]. These settings make body variation a central modeling object. The structural question here is complementary: when data are sparse and correspondences are not instance-level, action labels and target examples may specify what action should appear on the target body without specifying which source motion intention selected it.

Identifiability and distributional transport. Our formal argument is closest in spirit to identifiability limits in representation learning, where marginal fit alone does not recover latent structure without suitable inductive bias or auxiliary variables [Locatello et al., 2019, Khemakhem et al., 2020]. Distributional transport gives a second comparison point, since Schrödinger bridge objectives can learn a coupling between marginals without observing instance-level correspondences [Bortoli et al., 2021, Kim et al., 2024]. In cross-skeleton retargeting, however, the missing object is not merely a transport between two marginal motion distributions. It is the relative alignment that tells a target decoder how to interpret the source instance.

Motion libraries and retrieval comparators. Motion-library methods preserve realized target-side motion snippets by construction, and this property connects classical motion graphs to modern sparse-correspondence transfer [Kovar et al., 2002, Chen et al., 2025b]. This makes retrieval useful as a diagnostic foil: it can expose what action-level evidence accepts without claiming to synthesize a new source-conditioned motion. We use this distinction later to define a target-library comparator whose role is diagnostic rather than generative. Appendix A gives additional context for these connections.

3 The Source-Conditioned Map

Retargeting concerns a map that takes each source clip to its own target motion, not only a distribution of plausible target motions. Let $\mathcal{X}_s$ denote the motion space of skeleton s , with clips annotated by an action label c ; this notation records that clips on different bodies occupy different kinematic domains. A retargeting claim asserts a family of maps $T_{a \rightarrow b} : \mathcal{X}_a \rightarrow \mathcal{X}_b$ such that a source clip x_a on skeleton a is carried to skeleton b while preserving the action, timing, and motion intention of that clip. This is a stronger object than a conditional generator for plausible motions on b given the label c : the map must change when the source instance changes, even when the action and target skeleton are held fixed.

Sparse heterogeneous motion libraries provide much weaker evidence than this map requires. In the Truebones zoo used in our evaluation [Truebones Motions, 2025, Gat et al., 2025], the dataset contains 70 skeletons and 616 clips across 90 exact actions, yielding 6,300 possible skeleton-action cells. Only 413 of these cells are occupied, a fill rate of 6.6%, and the median occupied cell contains one clip. Thus a learner often sees that a skeleton can perform an action, but not multiple within-action

instances from which the geometry of motion intention could be inferred, much less a dense set of source-target correspondences specifying how the same instance should appear on a different body.

This sparsity matters because source-instance evidence is relational. To ask whether a map preserves the source instance, one must be able to compare several source clips that share a source skeleton and action, generate the corresponding motions on the same target skeleton, and then ask whether the source-side geometry remains visible after retargeting. Truebones contains only 49 such source groups, a source skeleton and action with at least three clips. Pairing each group with every other skeleton that performs the same action yields 1,891 source-target-action triples; these add target bodies but no new source motions, so all of them draw on the same 171 source clips, spread over eight actions in which attack and idle account for 94 percent of the triples. The structural question is therefore not whether action labels are useful, since they plainly are, but whether this sparse evidence selects a source-conditioned map rather than a target-action habit.

The distinction fixes the burden for the rest of the paper. Action-label evidence constrains which target motions are admissible for a requested action, whereas source-instance evidence constrains how differences among same-action source clips should reappear after the body changes. The former can support recognizable cross-skeleton generation; only the latter can identify the map that a retargeting claim asserts. Section 4 makes this gap precise by showing how standard objectives can leave multiple source-conditioned maps consistent with the same observed evidence.

4 Structural Non-Identifiability

A training objective can reward plausible target motions without caring which source-conditioned map produced them. This section makes the previous ambiguity structural by isolating two objective-level mechanisms: per-skeleton marginal matching leaves a relative latent coordinate freedom between bodies, while cell-paired squared-error regression under random target-cell pairing selects a target-action prototype. Both mechanisms preserve the evidence consumed by the objective while changing, or erasing, the source-instance dependence that retargeting requires.

The first mechanism appears whenever skeleton-specific encoders and decoders are coupled only through per-skeleton latent marginals. Let P_s be the motion distribution for skeleton s , let $\mu_s = (E_s)_\# P_s$ be the latent marginal, and write a shared-latent objective schematically as

$$L(E, D) = \sum_s L_s(D_s \circ E_s; P_s) + \Omega(\mu_1, \dots, \mu_K), \quad (1)$$

where Ω depends only on the per-skeleton latent marginals. For measurable invertible latent transformations $g = (g_1, \dots, g_K)$ with measurable inverses, define

$$G_\Omega(\mu) = \{g : \Omega((g_1)_\# \mu_1, \dots, (g_K)_\# \mu_K) = \Omega(\mu_1, \dots, \mu_K)\}. \quad (2)$$

For $g \in G_\Omega(\mu)$, set $E'_s = g_s \circ E_s$ and $D'_s = D_s \circ g_s^{-1}$. The relative transformations $g_b^{-1} \circ g_a$ are invisible to the marginal objective, but they are exactly the transformations that determine how a source latent is interpreted by a target decoder.

Theorem 1 (Gauge non-identifiability). *Let L be the objective in Eq. (1) with Ω depending only on per-skeleton latent marginals, and let $g \in G_\Omega(\mu)$ as in Eq. (2). Then $L(E', D') = L(E, D)$. If, in addition, there exist a, b such that $D_b \circ g_b^{-1} \circ g_a \neq D_b$ on a set of positive μ_a -measure, the induced retargeting maps differ: $T'_{a \rightarrow b} = D_b \circ g_b^{-1} \circ g_a \circ E_a \neq T_{a \rightarrow b} = D_b \circ E_a$ on a set of positive P_a -measure. $T_{a \rightarrow b}$ is determined by L only up to the relative gauge $g_b^{-1} \circ g_a$.*

This theorem is narrower than a generic impossibility result for disentanglement [Locatello et al., 2019] or nonlinear independent component analysis. It does not say that no auxiliary information can identify a representation. It says that the objective class above, when supplied with sparse skeleton-and-action evidence, does not select a relative latent alignment between skeletons. Identifiability under iVAE-style auxiliary variables [Khemakhem et al., 2020] requires the auxiliary distribution to induce full-rank variation in the prior sufficient statistics; on Truebones, where only 413 skeleton-action cells out of 6,300 are occupied and the median occupied cell contains one clip, skeleton-and-action metadata do not provide the empirical support needed to invoke that condition.

For Gaussian-aligned regularizers, the invariance contains $O(d)^K$; with $K = 70$ skeletons and the $d = 32$ latent dimension used for the finite-dimensional lower-bound calculation, quotienting the

diagonal $O(d)$ subgroup, one common orthogonal factor applied to all skeletons, still leaves a lower bound of 34,224 continuous gauge degrees of freedom. This is not a count of distinct realized maps, because an observable retargeting ambiguity requires *decoder non-degeneracy*: the decoder must discriminate at least some directions on the gauge orbit. That condition is architecture-dependent. The checks in Appendix D record a small Kullback-Leibler (KL)-regularized VAE whose decoder nearly absorbs the orbit, with a gauge-to-noise ratio of 1.11, and a separate AnyTop hidden-state sweep showing that a transformer diffusion decoder for arbitrary skeleton topology [Gat et al., 2025] can make an orthogonal perturbation orbit visible after decoding. The AnyTop sweep is not the source of the $d = 32$ count; it is a decoder-observability boundary check. Orbit observability is nevertheless distinct from gauge-element selection: a decoder can distinguish rotation directions without the objective selecting which rotation aligns source and target latents. The main point is therefore not that every latent coordinate change is observable, but that the objective does not identify the relative latent alignment needed for a source-preserving map. In this sense Theorem 1 is a statement about identifiability: the objective admits many source-conditioned maps consistent with the same evidence. It does not predict that a trained model’s outputs collapse, nor where a method falls on SIF.

The second mechanism arises when sparse paired evidence is treated as if it were instance-level supervision. Consider a predictor trained with squared error from a source clip x_a and target skeleton b to a target motion x_b . If the target is drawn uniformly from the target skeleton and source-action cell rather than paired to the same motion instance, the risk is

$$L_{\text{paired}}(f) = \mathbb{E} [\|f(x_a, b) - x_b\|_2^2]. \quad (3)$$

The Bayes-optimal squared-error predictor is then the conditional mean of the target cell [Hastie et al., 2009]; in this regime, the conditional mean is a target-action prototype rather than a source-conditioned motion.

Proposition 1.1 (Conditional-mean degeneration). *Assume x_b is square-integrable, f ranges over measurable functions, and the empirical pairing draws targets uniformly at random from the (skeleton, action) cell $B(b, c) = \{x : (\text{skel}(x), \text{action}(x)) = (b, c)\}$ with $c = \text{action}(x_a)$. Then the (almost surely unique) minimiser of L_{paired} is $f^*(x_a, b) = |B(b, c)|^{-1} \sum_{x'_b \in B(b, c)} x'_b$, the empirical cell mean: a single fixed prototype per (target skeleton, source action), independent of the specific source clip.*

The proposition is deliberately scoped. It applies to cell-paired squared-error regression under random target-cell pairing, not to true instance-level pairs, which would supply exactly the correspondence the sparse regime lacks. Section 5 tests this boundary on paired human-to-robot data: under random target-cell pairing a model’s outputs collapse toward the cell mean, while the same model trained on true pairs recovers the source-instance relation. It also assumes the target cell is nonempty and that motions are represented in the Euclidean vector space on which the squared-error loss is evaluated. The controlled ladder in Figure 2 shows the same mechanism in a setting where the correct source-conditioned transport T^* is known; the oracle variance is the within-cell variance of $T^*(x_a)$ over held-out source clips. As paired support per cell ranges from $M = 2$ to $M = 50$, the squared-error regressor trained with random same-cell targets keeps only about 2% of that oracle within-cell output variance. No evaluated Truebones method exactly instantiates this random cell-pairing model. The closest real-data echo is AL-Flow, a label-conditional flow comparator that receives cluster and exact-action labels but no source motion. The latent-level diagnostic of Section 5 (Table 16) places AL-Flow at -0.098 , indicating that source-instance information can be lost before decoding when the training signal approximates cell-level pairing rather than instance-level correspondence. Appendix B places each evaluated method relative to Theorem 1 and Proposition 1.1; for these methods, the collapse of output variation reported in Section 5 is an empirical observation rather than a prediction of either result. Appendix C contains the formal proofs: the gauge theorem in Appendix C.1, the conditional-mean proposition in Appendix C.2, and the retrieval corollary in Appendix C.3.

These two mechanisms also explain why retrieval comparators can be useful diagnostics without identifying the retargeting map. The following formal check records a boundary condition: a rule whose ranking depends only on features computed from joint positions, rather than from a learned shared latent, is outside the gauge variables of the objective above. It can therefore characterize the source-blind floor without selecting the source-conditioned map.

Corollary 1.1 (Retrieval is gauge-invariant). *Let $r : X_a \times \{b\} \rightarrow L_b$ be a retrieval rule that maps a source to a clip in the target-skeleton library by a function of a feature ϕ that is itself gauge-invariant, for example kinematic descriptors computed directly from joint positions rather than from a learned*

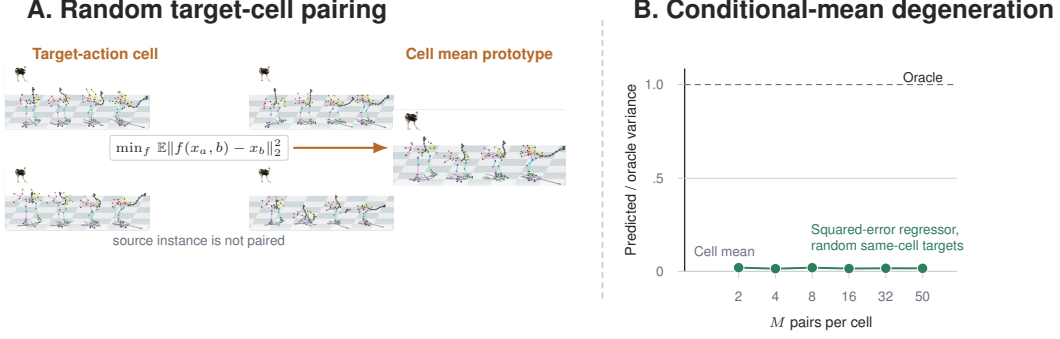


Figure 2: Cell-paired squared-error regression collapses toward the target-cell prototype. Panel A illustrates random target-cell pairing: several admissible target clips define a cell, but the squared-error optimum is the cell mean prototype rather than a source-conditioned motion. Panel B reports the controlled ladder with known source-conditioned transport T^* ; from $M = 2$ to $M = 50$ pairs per cell, predicted output variance stays at about 2% of the oracle within-cell variance $\text{Var}(T^*(x_a))$. The shaded band is the three-seed standard deviation.

latent. Then r is invariant under the gauge group G_Ω of any shared-latent objective: replacing E_a with $g_a \circ E_a$ leaves $r(x_a, b)$ unchanged. Furthermore $r(x_a, b)$ is a single existing clip rather than an average, so the conditional-mean operator of Prop. 1.1 does not apply.

The corollary describes a measurement, not a method to recommend. Its invariance holds because the score is computed from the motion itself rather than from learned latent coordinates. We later instantiate this measurement role through label-only baselines and ANCHOR, our deterministic target-library retrieval comparator. These comparators can mark what action-level evidence accepts at the source-blind floor precisely because they avoid the two objective mechanisms analyzed here. They do not identify the retargeting map; they expose why standard evidence alone cannot.

5 Source-Instance Fidelity and Empirical Evidence

What the action-level score leaves out is whether differences among source clips survive retargeting. A source-conditioned map should not merely produce a plausible target motion; after the source skeleton, target skeleton, and action are fixed, it should preserve the relative geometry among source clips. We therefore define *Source-Instance Fidelity* (SIF) on each eligible triple (a, b, c) . For source clips $x_i, x_j \in \mathcal{X}_a$ and generated target motions $y_i, y_j \in \mathcal{X}_b$, each pair is cropped to its common duration and put into a common comparison frame by least-squares Procrustes superposition, which removes global translation, rotation, and isotropic scale [Goodall, 1991]. Let $d_{\text{src}}(i, j)$ and $d_{\text{out}}(i, j)$ be the aligned per-frame discrepancies on the source and output sides. Using Pearson’s product-moment correlation [Pearson, 1896], we define

$$\rho_{\text{SIF}} = \text{CORR}(\{d_{\text{src}}(i, j)\}_{i < j}, \{d_{\text{out}}(i, j)\}_{i < j}), \quad (4)$$

within each source-target-action triple and report the macro-average over triples. Because the same source clips recur across triples, confidence intervals resample whole source skeletons. We score every method twice: in raw form, as above, and in length-controlled form, where every motion is first resampled to 64 frames so that clip duration carries no information. Beside the correlation we report the output variation, the average distance between outputs divided by the average distance between their source clips; it is near one when outputs vary as much as their sources and zero when they collapse to a single motion.

SIF asks a different question from action retrieval. A generator can recover the right target action by returning a target-action habit, in which case the action-level score may be positive while ρ_{SIF} remains near zero. A source-preserving map should instead make larger source-side changes appear as larger target-side changes after target skeleton and action are fixed. For failure-mode analysis, we also compute *latent Source-Instance Fidelity* (L-SIF) before decoding, replacing output distances with Frobenius distances between the corresponding target latents [Golub and Van Loan, 1996]. This locates whether source-instance information is absent in the latent representation or erased by the decoder.

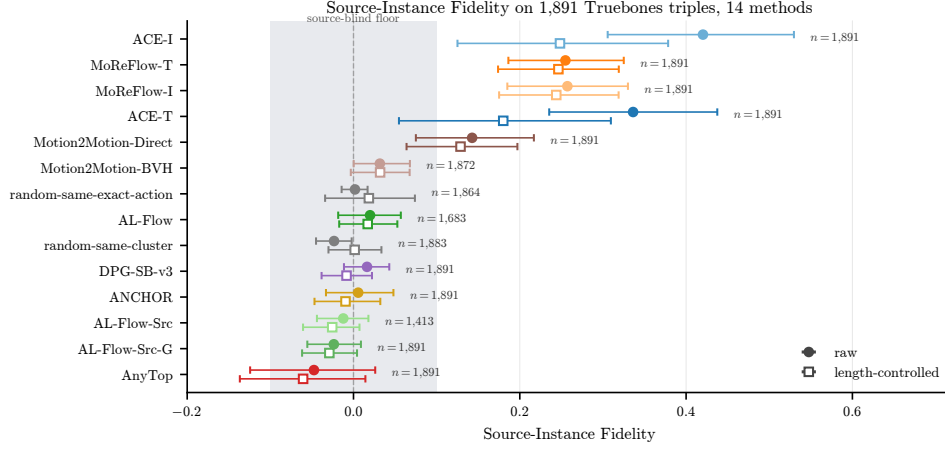


Figure 3: Source-Instance Fidelity for the fourteen evaluated methods on the 1,891 Truebones triples. Filled circles use raw scoring and open squares length-controlled scoring; intervals are 95% bootstrap intervals that resample whole source skeletons, and n is the number of triples a method supports. The shaded band marks $|\rho_{\text{SIF}}| < 0.10$. Eight methods do not stand out from shuffled pairings under either scoring, and Motion2Motion-BVH stays near zero.

The synthetic calibration fixes the scale of the diagnostic. In the controlled 2×2 setting, the source-conditioned transport T^* is known for every source clip, providing the instance-level correspondence absent from the real corpus. Appendix F shows that the oracle sits near one, while two source-blind references are centered near zero with upper bootstrap bounds below $+0.06$. On real data, a generator given no source at generation, with output length fixed, scores exactly zero (Appendix I). To decide whether a method sits at this floor, we use a shuffle test: within each triple, outputs are reassigned at random among the source clips, the true pairing included. A method is at the source-blind floor when its SIF does not stand out from these shuffles under both raw and length-controlled scoring, and near the floor when it does but stays inside the $|\rho_{\text{SIF}}| < 0.10$ band set by the synthetic references.

On the real corpus, SIF must be reported with the support that sparse data allow. We evaluate every method on the 1,891 triples of Section 3; methods that cannot produce an output for a triple are scored on the remainder, which never falls below 1,413 triples. The original evaluation, which pairs each source group with a single target (49 triples), is reported in Appendix E. The suffixes T and I mark models trained on all 70 skeletons or on 60, with the other 10 held out as targets. The comparison is organized by evidentiary role rather than by rank: published generative objectives are evaluated beside rows that isolate sparse correspondence, supervision-matched labels, latent transport, target-library retrieval, and label-only sampling [Gat et al., 2025, Kim et al., 2025, Li et al., 2023, Chen et al., 2025b, Bortoli et al., 2021, Kim et al., 2024]. Appendix B records source-conditioning adaptations and split conventions.

ANCHOR occupies a distinct diagnostic role: it is not a retargeting generator. Given a source clip, target skeleton, and action request, it predicts a coarse source action cluster from motion-space descriptors and retrieves an existing target-library clip using that prediction, exact action metadata, and direct kinematic descriptors. Because this rule sits outside the learned latent objectives analyzed in Section 4, it can show when action-level evidence and source-conditioned transfer separate. The scoring rule appears in Appendix G.

The main empirical pattern is that action-consistent generation usually does not preserve source-instance geometry. Figure 3 shows that nine of the fourteen evaluated methods sit at or near the source-blind floor under both raw and length-controlled scoring: AnyTop, the three AL-Flow variants, DPG-SB-v3, ANCHOR, and both random label-only references do not stand out from shuffled pairings, and Motion2Motion-BVH stays near zero ($+0.032$). Five methods rise above the floor. Motion2Motion-Direct keeps a modest correlation ($+0.14$ raw, $+0.13$ length-controlled). MoReFlow-T and MoReFlow-I remain above the floor at about $+0.25$ under both scorings, and their outputs keep 16 to 21 percent of the source variation. MoReFlow trains on source-target pairs matched by motion descriptors computed from joint positions, a coarse form of the correspondence evidence the

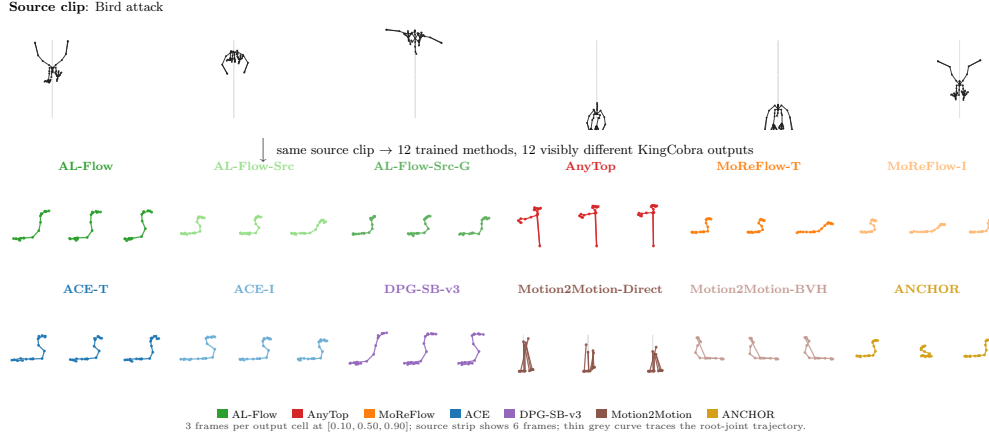


Figure 4: Qualitative non-identifiability panorama. A single Bird $\rightarrow$ KingCobra attack query is shown across evaluated methods. All rows receive the same source clip, target skeleton, and action request, yet the generated target motions differ visibly across methods. The figure is qualitative evidence for the ambiguity isolated by SIF: action-consistent target motion can be produced without identifying a unique source-conditioned map.

sparse regime lacks; we did not isolate this matching step. ACE-T and ACE-I score higher in raw form (+0.34 and +0.42) but drop under length control (+0.18 and +0.25), and their outputs vary by only 2 to 3 percent of the source variation in raw scoring and by less than 1 percent when the length is fixed at generation. The result is consistent with the structural claim in Section 4: action-level evidence can be present while the source-conditioned map remains unidentified. Appendix E reports the corresponding support, bootstrap, and latent-SIF checks.

The qualitative panorama makes the same ambiguity visible. In Figure 4, a single Bird $\rightarrow$ KingCobra attack query is held fixed while the evaluated methods produce visibly different target motions. The figure is not a substitute for SIF, but it shows why a visual action judgment alone cannot identify the source-conditioned map: many outputs can look action-consistent while corresponding to different maps.

The panorama does not ask the reader to choose a visually best method, nor does it treat a single attractive output as evidence of transfer. It fixes the query and shows the range of outputs that remain possible once the action is recognizable. SIF then supplies the missing relational test: if the source clip changes within the same source-target-action cell, the target output should change in the corresponding way rather than merely remain a plausible KingCobra attack.

Action-level AUC rewards any output with a recognizable target action, including outputs that SIF places at the source-blind floor. Table 1 gives the numerical audit, and Figure 5 visualizes the same comparison. The random-same-cluster reference chooses a random target clip with the same coarse action cluster, while random-same-exact-action chooses a random target clip with the exact action label whenever such a clip exists. These rules have no access to a source-conditioned retargeting map, yet random-same-cluster matches ANCHOR within ± 0.012 on both the overall and held-out splits. The exact-action variant exceeds ANCHOR by +0.126 overall and +0.242 held-out, but only on the 221/600 overall and 57/200 held-out coverage where exact-action positives exist in the target library. The qualifier is part of the claim: strong action-level evidence can be generated by label availability alone, without identifying the map. Appendix H gives the broader enumeration and action-level comparison tables behind this audit.

The label-only audit changes how the positive SIF rows should be read. Once action-level AUC can be reproduced by references with no source-conditioned map, the question is not whether a method also scores well on action recovery, but whether it contains independent motion-space evidence that preserves the source instance after the target skeleton and action are fixed. The rows above the floor answer this only in part. ACE retains a relational signal that enters through its source input: with the source removed and the output length fixed at generation, its SIF falls from +0.29 to exactly zero

Table 1: Label-only retrieval can match or exceed ANCHOR’s action-level AUC without solving the retargeting problem. AUC is measured under the same cluster-tier Procrustes protocol used for ANCHOR. The exact-action reference has restricted coverage, so its larger gaps are only defined on the rows where the target library contains an exact-action candidate.

Method	Overall $n/600$	Held-out $n/200$	Overall AUC	Held-out AUC
ANCHOR	600	200	0.757	0.681
random-same-cluster	570	177	0.769 (+0.012)	0.678 (-0.003)
random-same-exact-action	221	57	0.883 (+0.126)	0.923 (+0.242)

Parentetical Δ values are method AUC minus ANCHOR. random-same-cluster matches ANCHOR within ± 0.012 on both splits. random-same-exact-action exceeds ANCHOR by +0.126 overall and +0.242 held-out, but only on the 221/600 and 57/200 coverage where exact-action positives exist; the coverage qualifier travels with the gap.

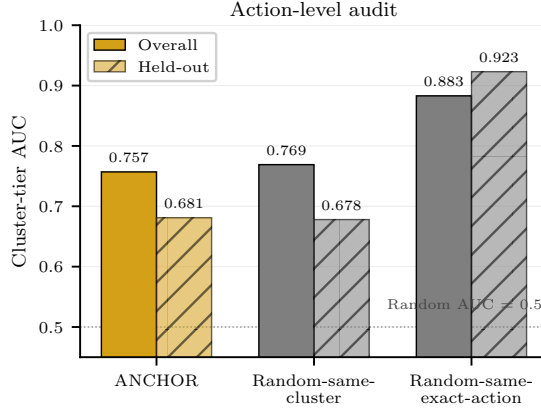


Figure 5: Action-level AUC cannot distinguish source-conditioned transport from label-based retrieval. ANCHOR retrieves from the target library using action information and motion descriptors, while the two random references use only label constraints. Their AUC values remain in the same action-level regime, showing that the standard score can accept target-action recovery even when source-instance transfer has not been established. The exact-action reference is coverage-limited to 221/600 overall and 57/200 held-out queries.

(Appendix I). Its outputs, however, vary little with the source, and under length control its correlation cannot be told apart from MoReFlow’s on the same triples.

The ACE controls in Appendix I also test where this signal comes from. Removing the motion-space adversarial loss, in a comparison matched in training data and seeds with the output length fixed at generation, leaves SIF unchanged (+0.33 with and without the loss on the 1,891 triples, three seeds each), so the adversarial constraint does not create the signal. The nearest-neighbor leakage check in Table 26 gives the complementary control: 76 to 79 percent of ACE outputs are closer, under coarse kinematic features, to a target-skeleton training clip than to the source clip, which does not support a direct source-copying explanation. The structural lesson is therefore specific: an auxiliary objective can leave a partial source-instance signal without selecting a source-preserving map, and action-level AUC by itself cannot tell the two apart.

The same pattern holds where true correspondences exist. We built two paired human-to-robot evaluations with the SIF protocol unchanged (Appendix J). In the first, human motion clips from the BONES-SEED dataset come with their retargeted counterparts on the Unitree G1 humanoid (90 action groups, 355 clips). In the second, we retargeted clips from the LAFAN1 human motion dataset to six humanoid robots with the GMR inverse-kinematics tool, so every clip has a counterpart on every robot. In both, we trained one model in three ways: with an unpaired objective of the Theorem 1 class, with random same-action pairing as in Proposition 1.1 (the averaging objective), and on the true pairs. Table 2 reports the G1 results on identical test items. A randomly chosen same-action clip is recognized as the right action yet sits at the floor, the two theorem objectives lose most of the source variation, and the model trained on true pairs recovers both. Across the six robots, the averaging

Table 2: Human-to-G1 evaluation on identical test items (90 action groups, 355 clips). SIF uses raw scoring; variation is the median ratio of output spread to source spread; realistic is the share of outputs whose nearest real G1 pose lies within the 95th percentile of held-out real motion.

	SIF	Action AUC	Variation	Realistic
True retargeted counterpart	+0.960	0.980	0.52	97.2%
Random same-action clip	−0.008	0.978	0.41	97.5%
Unpaired objective (Theorem 1 class)	+0.203	0.499	0.001	100.0%
Averaging objective (Proposition 1.1 class)	+0.379	0.820	0.05	53.0%
Model trained on true pairs	+0.900	0.967	0.52	94.6%

objective collapsed in all 24 evaluations and the unpaired objective in 22, while the true-pair model reached correlations of 0.77 to 0.99 in all 24. These robots are humanoids, their pairs come from retargeting tools rather than animators, and the sparsity is created by us.

6 Limitations

SIF asks whether outputs are ordered like their sources. It does not measure how much they differ, so a method can score well while its outputs barely change; we therefore report output variation beside it. For a single group of three clips the score is noisy, and it becomes reliable only as an average over many groups (Appendix E). A positive score can also appear without any transfer of source motion. In our study, outputs that copied the length of their source clip, a source latent left available at generation, and random draws from small pools of same-action clips each produced one; length-controlled scoring and the controls in Section 5 and Appendix I separate these effects from source content. The animal evidence comes from a single dataset, and the enlarged evaluation adds target bodies rather than new source motions. The robot studies extend the picture to humanoids, but their paired motions come from retargeting tools rather than animators, and their sparsity is created by us. Several evaluated methods are also adaptations of published models to this data, and our theorems cover specific training objectives rather than every method we evaluate.

7 Conclusion

Cross-skeleton retargeting can look successful under action-level evaluation even when the source-conditioned map has not been identified. We traced this to two structural mechanisms. Under *gauge non-identifiability*, matching each body’s motions only in distribution leaves the latent spaces of different bodies free to be transformed relative to one another, so many maps fit the data. Under *conditional-mean degeneration*, pairing clips only by action drives squared-error training to an average target motion instead of the motion intention of the source clip. Source-Instance Fidelity makes the missing evidence measurable by asking whether outputs differ the way their source clips do.

In sparse animal motion data, outputs that show the right action often sit at the source-blind floor, and label-only references show that action-level AUC can stay high without any map being selected. Methods that rise above the floor, including ACE, retain only a partial relational signal, while on paired human-to-robot data a model trained on true pairs recovers the source-instance relation. Progress in retargeting should therefore be judged by whether an objective improves action recovery, source-conditioned transport, or both. The open question is which correspondence evidence or auxiliary objectives can select a source-preserving map from the many maps that fit the same data.

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A Extended Related Work

Cross-skeleton retargeting sits between character animation, heterogeneous motion generation, and representation learning. Earlier retargeting work established that motion can be shared across bodies by making skeletal structure explicit, by learning motion spaces intended to be less dependent on topology, or by conditioning sequence models on skeleton context [Aberman et al., 2020, Lee et al., 2023, Wang et al., 2024]. The generative setting studied here inherits this ambition, but places it in sparse heterogeneous libraries where dense source-target correspondences are absent [Gat et al., 2025, Kim et al., 2025, Li et al., 2023, Chen et al., 2025b]. Our distinction is not that these methods fail to synthesize plausible target motion. It is that standard evidence can accept a target motion whose action is correct while the source-conditioned map remains unidentified.

Table 3: Truebones zoo regime. SIF is evaluated on source groups with at least three clips; the main evaluation pairs each group with every target skeleton that performs the same action, and the original evaluation pairs it with a single target.

Quantity	Value
Skeletons	70
Total clips	616
Coarse clusters	10
Exact actions	90
Skeleton-action cells	6,300 (70×90)
Occupied cells	413 (6.6%)
Median clips per occupied cell	1
Source groups with at least three clips	49 (171 source clips)
SIF triples, every eligible target (main evaluation)	1,891 (6,611 queries)
SIF triples, one target per group (original evaluation)	49 (171 queries)
Original cross-method intersection	37 triples (130 queries)

Recent work on broader object and animal motion further clarifies why this identification question matters. Together, these papers move motion synthesis away from one canonical human body and toward settings in which body identity, morphology, and topology become part of the generative problem itself [Raab et al., 2024, Lee et al., 2025, Egan et al., 2024, Chen et al., 2025a]. These works broaden the range of bodies that motion models must handle. The structural question in this paper is complementary: in sparse heterogeneous data, which evidence selects the motion intention carried from the source clip rather than a target-side action habit?

The formal argument is closest in spirit to identifiability limits for unsupervised representation learning. Locatello et al. [Locatello et al., 2019] show that unsupervised disentanglement cannot be guaranteed without inductive bias, while the iVAE analysis of Khemakhem et al. [Khemakhem et al., 2020] shows how suitable auxiliary variables can make nonlinear independent components identifiable. Our gauge theorem is narrower than either statement. It does not rule out identification with stronger side information; it shows that per-skeleton marginal matching does not select the relative latent alignment required by a source-preserving retargeting map.

Retrieval and motion-library methods provide a useful foil because they operate directly in motion space. Motion graphs [Kovar et al., 2002] already exploit the idea that a library can preserve physically realized motion snippets, while sparse-correspondence retrieval shows how such libraries can be used when source and target bodies do not share topology [Chen et al., 2025b]. These methods motivate the diagnostic comparison used in the main text: retrieval can characterize what action-level evidence accepts, but by itself it does not identify the source-conditioned retargeting map.

Schrödinger bridge and flow objectives offer another route through the same space of ambiguities. Diffusion Schrödinger bridges [Bortoli et al., 2021] and unpaired neural Schrödinger bridges [Kim et al., 2024] motivate transport between distributions, and flow matching is one way modern retargeting methods instantiate this distributional view [Kim et al., 2025]. In the sparse cross-skeleton setting, distributional transport alone does not remove the relative-gauge ambiguity. This is why the paper separates action-level transport evidence from the source-instance evidence measured by SIF.

B Data Regime and Method Roster

The structural limits studied in the main paper arise in a sparse heterogeneous motion library. Table 3 records the Truebones regime used throughout the experiments. The important fact is not merely that the library is small, but that most skeleton-action cells are empty and most occupied cells contain a single clip, making instance-level correspondence rare even when action labels are available.

The evaluated roster in Table 4 separates published generative models, supervision-matched comparators, the deterministic ANCHOR comparator, and source-blind random references. The public method names in this table are the only names used for result interpretation.

Table 4: Method roster. Train-skeleton counts distinguish transductive evaluation on all skeletons from inductive evaluation with held-out target skeletons.

Method	Family	Train skeletons
AnyTop	source-conditioned diffusion	70
ACE-T	motion-space adversarial	70
ACE-I	motion-space adversarial	60
MoReFlow-T	flow matching	70
MoReFlow-I	flow matching	60
AL-Flow	label-conditional flow, no source motion	60
AL-Flow-Src	label-conditional flow + source motion + skeleton id	60
AL-Flow-Src-G	label-conditional flow + source motion + skeleton graph	60
Motion2Motion-Direct	patch retrieval and blending (training-free)	60
Motion2Motion-BVH	patch retrieval and blending (training-free)	60
DPG-SB-v3	latent Schrödinger bridge	60
ANCHOR	retrieval comparator	60
random-same-cluster	trivial baseline	—
random-same-exact-action	trivial baseline	—

The AnyTop row uses the source-conditioned extension required by SIF; the original generator is unconditional with respect to a source clip. AL-Flow denotes the label-conditional flow family. ANCHOR is a deterministic retrieval comparator, not a retargeting solution. DPG-SB-v3 is evaluated in its latent-bridge form without decoded-motion penalties (see Appendix L).

Table 5: Relation of each evaluated method to the formal results.

Method	Relation to the formal results
AnyTop	closest to Theorem 1; adds rank and view-consistency terms
AL-Flow, AL-Flow-Src, AL-Flow-Src-G	closest to Proposition 1.1; trained with flow matching rather than squared error
MoReFlow-T, MoReFlow-I	partly within Theorem 1; adds pairing by matched motion descriptors and a descriptor loss
DPG-SB-v3	partly within Theorem 1; adds adversarial and cycle terms in the latent space
ACE-T, ACE-I	outside both classes; adversarial and source-feature losses
ANCHOR, random-same-cluster, random-same-exact-action	retrieval rules with no trained objective (Corollary 1.1)
Motion2Motion-Direct, Motion2Motion-BVH	patch retrieval and blending with no trained objective

Source-conditioning adaptations. Two rows require additional care because their original public form does not match the SIF input contract. AnyTop is unconditional with respect to a source clip, so the evaluated row adds a self-supervised source-motion encoder with four learned motion queries and injects the resulting latent through cross-attention into the target-skeleton decoder; training remains unpaired self-reconstruction on Truebones. Motion2Motion is a sparse-correspondence retrieval-and-blending procedure rather than a learned generator, so we report both Motion2Motion-Direct, a reimplement on the 13-channel representation, and Motion2Motion-BVH, the official BVH implementation, using auto-heuristic correspondences and excluding evaluator candidate clips from the target example pool. These details affect only how each row receives source information; SIF is computed by the same source-target-action protocol across rows.

Relation to the theory. Table 5 places each evaluated method relative to the two formal results. Theorem 1 covers objectives that only match each skeleton’s motion distribution through a shared Gaussian latent, without paired examples. Proposition 1.1 covers squared-error training against a randomly chosen same-action target. No evaluated method instantiates either class exactly, and the placement describes the structure of each objective rather than a predicted SIF value.

C Proofs and Formal Details

C.1 Proof of Theorem 1

Proof. For each skeleton s and motion $x \in \mathcal{X}_s$, the reparameterized encoder-decoder pair satisfies

$$D'_s(E'_s(x)) = D_s(g_s^{-1}(g_s(E_s(x)))) = D_s(E_s(x)),$$

because g_s is a measurable bijection. The per-skeleton reconstruction term $L_s(D_s \circ E_s; P_s)$ is therefore unchanged pointwise.

The latent marginal induced by the reparameterized encoder is $(E'_s)_\# P_s = (g_s \circ E_s)_\# P_s = (g_s)_\# \mu_s$. Since $g \in G_\Omega(\mu)$, the marginal regularizer also remains unchanged:

$$\Omega((g_1)_\# \mu_1, \dots, (g_K)_\# \mu_K) = \Omega(\mu_1, \dots, \mu_K).$$

Combining the unchanged reconstruction terms with the unchanged regularizer gives $L(E', D') = L(E, D)$.

For the retargeting map, define

$$A = \{z : D_b(g_b^{-1}(g_a(z))) \neq D_b(z)\}.$$

By the decoder non-degeneracy hypothesis in Theorem 1, this set has positive μ_a -measure for some pair a, b . Let $B = E_a^{-1}(A) \subseteq \mathcal{X}_a$. Since $(E_a)_\# P_a = \mu_a$, $P_a(B) = \mu_a(A) > 0$. For every $x \in B$,

$$T'_{a \rightarrow b}(x) = D_b(g_b^{-1}(g_a(E_a(x)))) \neq D_b(E_a(x)) = T_{a \rightarrow b}(x),$$

so the two retargeting maps differ on a set of positive P_a -measure. The objective therefore determines the map only up to the relative gauge $g_b^{-1} \circ g_a$. $\square$

Corollary 1.2 (Orthogonal-gauge lower bound under exact Gaussian alignment). *Suppose $\Omega(\mu_1, \dots, \mu_K) = \sum_s \text{KL}(\mu_s \| \mathcal{N}(0, I_d))$, or any rotation-invariant function of the marginals, and each $\mu_s = \mathcal{N}(0, I_d)$. Then G_Ω contains $O(d)^K$, the product of orthogonal groups. The full product has $K d(d-1)/2$ continuous parameters. After quotienting the diagonal $O(d)$ subgroup, one common orthogonal factor applied to all skeletons, the relative gauge retains $(K-1)d(d-1)/2$ continuous degrees of freedom.*

Proof. If $R \in O(d)$ and $z \sim \mathcal{N}(0, I_d)$, then $Rz \sim \mathcal{N}(0, RR^\top) = \mathcal{N}(0, I_d)$. Thus every per-skeleton choice $(R_1, \dots, R_K) \in O(d)^K$ preserves all Gaussian-aligned marginals and belongs to G_Ω whenever Ω is rotation-invariant. Since $O(d)$ has dimension $d(d-1)/2$, the product group has $K d(d-1)/2$ continuous parameters. Quotienting the diagonal subgroup removes one $O(d)$ factor, because a common orthogonal transformation applied to all skeletons cancels in the relative transformation $g_b^{-1} \circ g_a$, leaving $(K-1)d(d-1)/2$ relative-gauge parameters. For the setting used in the paper, $K = 70$ and $d = 32$, giving $69 \cdot 32 \cdot 31/2 = 34,224$ relative-gauge degrees of freedom in this orthogonal subgroup. This is a lower bound on relative-gauge degrees of freedom, not by itself a count of distinct decoded retargeting maps; that latter count also depends on decoder non-degeneracy. $\square$

C.2 Proof of Proposition 1.1

Proof. Condition on a source clip x_a and target skeleton b , and write $c = \text{action}(x_a)$. Under the proposition's sampling rule, the target x_b is drawn uniformly from the finite cell $B(b, c) = \{x : (\text{skel}(x), \text{action}(x)) = (b, c)\}$, independently of the source instance once the action cell is fixed. For any proposed prediction $u = f(x_a, b)$, the sampling rule gives $x_b \perp x_a \mid (b, c)$, hence $\mathbb{E}[x_b \mid x_a, b] = \mathbb{E}[x_b \mid b, c]$. Therefore

$$\mathbb{E}\|u - x_b\|_2^2 = \|u - \mathbb{E}[x_b \mid b, c]\|_2^2 + \mathbb{E}\|x_b - \mathbb{E}[x_b \mid b, c]\|_2^2.$$

The second term is independent of u , so the finite-cell minimizer is unique; in the population version the same equality gives the almost surely unique conditional-mean predictor $u = \mathbb{E}[x_b \mid b, c]$. In the empirical finite-cell case, this conditional expectation is

$$|B(b, c)|^{-1} \sum_{x'_b \in B(b, c)} x'_b.$$

It depends on the target skeleton and source action, but not on the specific source clip. The Bayes predictor is therefore the target-action cell mean. $\square$

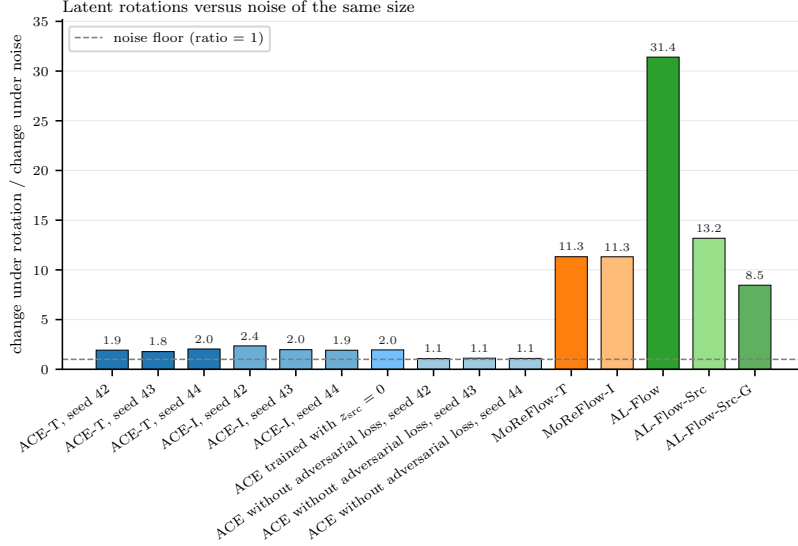


Figure 6: Decoder non-degeneracy check. Rotation-induced changes are compared with matched-magnitude noise in the latent space; ratios above one indicate that the decoder distinguishes at least some directions on the gauge orbit.

C.3 Proof of Corollary 1.1

Proof. The retrieval rule r depends on the source motion through a feature $\phi(x_a)$ that is computed directly from motion space and is invariant to the latent-gauge transformations in G_Ω . Replacing E_a by $g_a \circ E_a$ therefore changes no argument supplied to r , and the returned library clip is unchanged. Since $r(x_a, b)$ is a single existing target-skeleton clip rather than an average over target-cell motions, the conditional-mean operator in Proposition 1.1 is not the relevant limiting operation. $\square$

D Decoder Non-Degeneracy Boundary

The gauge theorem requires a decoder that can make at least part of the gauge orbit observable. In a small Kullback-Leibler (KL)-regularized VAE, random gauge rotations and matched-magnitude noise remain near the noise floor in decoded effects, with a gauge-to-noise ratio of 1.11. This is the boundary case in which the decoder nearly absorbs the orbit.

A separate AnyTop transformer diffusion hidden-state sweep tests decoder observability, not the $d = 32$ gauge-count model. Across hook layers, reverse timesteps, source motions, and target skeletons, all 72 tested cells, formed by four skeletons, two source clips, three hook layers, and three reverse timesteps, give rotation-induced output divergence above the matched-noise control, with a minimum gauge-to-noise ratio of 1.71. This check does not instantiate the 34,224-dimensional lower-bound calculation. It shows only that decoder observability can hold in a real transformer diffusion motion model, leaving gauge-element selection as the structural problem. Figure 6 and Table 6 report the same rotation-versus-noise check for the evaluated methods.

The cross-seed latent checks in Tables 7 and 8 give a complementary view. Orthogonal Procrustes alignment [Goodall, 1991] explains a substantial fraction of cross-seed latent variation, while effective rank separates rich latent variation from low-dimensional collapse. These checks defend the boundary of the theory: decoder observability can hold even when the objective does not select the relative gauge. Figures 7, 8, and 9 visualize the same distinction between latent alignment, representation richness, and source-instance fidelity.

Table 6: Rotation versus noise perturbations on the 130 queries of the 37-triple intersection. The ACE ablation with the motion-space adversarial loss removed is the boundary case near the noise floor, while the other evaluated methods make the orbit visible after decoding.

Method	Seed	Change under rotation	Change under noise	Ratio	Verdict
ACE-T (3 seeds)	42	0.240	0.213	1.93	non-degenerate
	43	0.219	0.183	1.78	non-degenerate
	44	0.209	0.171	2.04	non-degenerate
ACE-I (3 seeds)	42	0.121	0.089	2.35	non-degenerate
	43	0.128	0.071	1.98	non-degenerate
	44	0.141	0.087	1.92	non-degenerate
ACE trained with $z_{\text{src}} = 0$	42	0.117	0.083	1.95	non-degenerate
ACE without adversarial loss (3 seeds)	42	1.113	1.154	1.08	at noise floor
	43	0.810	0.849	1.12	at noise floor
	44	0.699	0.734	1.10	at noise floor
MoReFlow-T	–	0.113	0.017	11.33	non-degenerate (strong)
MoReFlow-I	–	0.107	0.016	11.32	non-degenerate (strong)
AL-Flow	–	0.080	0.008	31.40	non-degenerate (strong)
AL-Flow-Src	–	0.070	0.009	13.18	non-degenerate (strong)
AL-Flow-Src-G	–	0.050	0.008	8.46	non-degenerate (strong)

Each row reports the mean over the 130 queries of the 37-triple intersection, with 20 random $O(d)$ rotations and 20 matched-magnitude noise samples per query; the ratio is the mean of the per-query rotation-to-noise ratios. ACE without the adversarial loss is statistically indistinguishable from the noise floor (Wilcoxon $p > 0.4$ for all three seeds); every other method exceeds the noise floor at $p < 10^{-12}$.

Table 7: Orthogonal Procrustes alignment across seeds. The table reports how much latent variation can be explained by an $O(d)$ alignment across the SIF-intersection queries.

Group	Seed pair	Share explained by rotation	Cosine after rotation
ACE-T	42 vs 43	0.522	0.893
ACE-T	42 vs 44	0.703	0.900
ACE-T	43 vs 44	0.585	0.905
ACE-I	42 vs 43	0.689	0.935
ACE-I	42 vs 44	0.712	0.928
ACE-I	43 vs 44	0.775	0.941
ACE without adversarial loss	42 vs 43	0.482	0.866
ACE without adversarial loss	42 vs 44	0.428	0.847
ACE without adversarial loss	43 vs 44	0.753	0.881

Share explained by rotation is the fraction of cross-seed latent variance accounted for by the best $O(d)$ rotation (orthogonal Procrustes). *Cosine after rotation* is the mean per-row cosine similarity after that rotation. Both are computed on per-target-skeleton latents across the 130 queries of the 37-triple set, aggregated over 21 target skeletons.

E Source-Instance Fidelity Robustness

SIF is a diagnostic for whether source-side geometry remains visible after the target skeleton and action are fixed. This appendix gives the numbers behind Figure 3, the original evaluation on 49 triples, and checks on dependence, sample size, and generation noise.

Table 9 lists every method on the 1,891 triples of the main evaluation. Each source clip appears in many triples there, so the shuffle test moves all triples from the same source skeleton together, and the intervals resample whole source skeletons. Treating the triples as independent would overstate the precision. The last column applies the rule of Section 5.

Table 10 reports the original evaluation, which pairs each of the 49 source groups with a single target, so no source clip appears in two triples. The pattern matches the main evaluation, with one difference: Motion2Motion-Direct sits at the floor here and rises modestly above it only on the larger set. The random exact-action reference is just significant in raw scoring ($p = 0.049$) and returns to the floor under length control. Its raw signal comes from small candidate pools and clips cropped to their

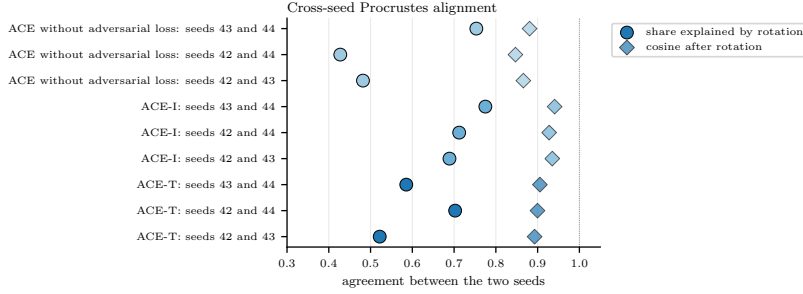


Figure 7: Cross-seed Procrustes alignment. High explained fractions indicate that latent spaces differ partly by approximately orthogonal transformations, matching the relative-gauge mechanism in Theorem 1.

Table 8: Effective-rank diagnostic on latent distributions. Higher effective rank indicates richer latent variation, but rank alone does not imply that the source-conditioned map has been identified.

Method	Sample shape	Spectral flatness	Effective rank / 256
ACE without adversarial loss (seed 42)	$3,376 \times 256$	0.0001	1.29
ACE-T (seed 42)	$3,376 \times 256$	0.0001	2.13
ACE-I (seed 42)	$3,376 \times 256$	0.0019	3.00
AL-Flow	$1,040 \times 256$	0.0066	6.80
MoReFlow-I	$3,823 \times 256$	0.0593	12.84
MoReFlow-T	$3,823 \times 256$	0.0631	14.99
AnyTop encoder z	$13,000 \times 256$	0.1614	57.73

Effective rank computed on the per-method latent distribution over the 130 queries of the 37-triple intersection. AnyTop’s encoder hidden state is an order of magnitude richer than the compressed latents shared by the MoReFlow-based models.

source’s length. Table 11 and Figure 10 give raw-scoring intervals on the 37 triples that all methods support, clustered by source skeleton, target skeleton, or action.

Table 12 compares ACE with MoReFlow on identical triples. In raw scoring ACE-I scores higher than MoReFlow-I. Under length control, neither ACE model differs from MoReFlow in either evaluation.

A single triple with three source clips gives a noisy estimate. To measure this, we took the triples with five source clips and recomputed SIF on every three-clip subset. Table 13 shows that the value for one triple moves with a standard deviation of about 0.6 across subsets. The average over randomly drawn sets of triples moves by about 0.12 for 25 triples and by about 0.02 for 500. Because the 1,891 triples reuse the same 171 source clips, the uncertainty reported for the main evaluation comes from the intervals that resample whole source skeletons (Table 9).

Methods that sample noise raise a further question: whether shared noise can create an apparent signal. Table 14 regenerates every stochastic method with three independent noise seeds, and once with a single noise sample shared within each triple. With independent noise, every stochastic method stays within 0.05 of zero on average. With shared noise, AnyTop and the source-fed AL-Flow variants rise to about +0.2. The outputs then differ only through their inputs, such as length, and SIF picks up that difference without any source content. All results in the paper use independent noise. MoReFlow is deterministic, so its score does not depend on noise.

The eligibility rule also matters for interpretation. Table 15 and Figure 11 show that SIF triples have at least three source clips by construction, but only a few triples provide more than three. Figure 12 checks that the main pattern is not a monotone artifact of output variation (the diversity ratio R).

Latent SIF (L-SIF) repeats the SIF computation before decoding, when a method exposes a latent representation on the relevant queries. Table 16 and Figure 13 show that latent preservation and decoded SIF can decouple. This motivates the failure-mode taxonomy used in the qualitative discussion: a method may collapse in latent space, collapse after decoding, or keep a positive output correlation while its outputs barely vary (collapsed variation).

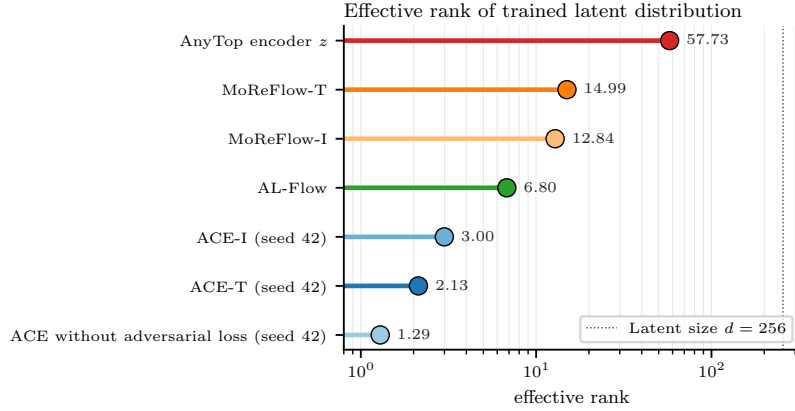


Figure 8: Effective rank across latent representations. The diagnostic separates representation richness from source-instance fidelity.

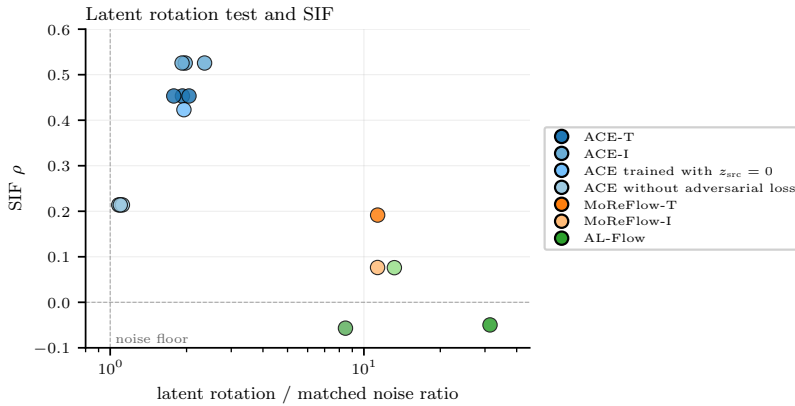


Figure 9: Decoder non-degeneracy and SIF are distinct. A method may expose latent perturbations after decoding while still lying near the source-blind floor under SIF.

F Synthetic Calibration

The synthetic 2×2 environment calibrates SIF in a setting where the source-conditioned transport is known. Figure 14 visualizes the three reference regimes used in Section 5: the oracle transport lies near one, while two source-blind references lie near zero. Table 17 records the corresponding numerical calibration.

The same environment also isolates conditional-mean degeneration. The cell-paired squared-error predictor receives random target samples from the correct target skeleton and action cell, but not instance-level pairs. As Table 18 shows, increasing the number of random cell-pairs does not recover the oracle within-cell variance.

G Deterministic Comparator and Motion Descriptors

ANCHOR is included to make the diagnostic boundary explicit. It is not a generator and does not synthesize a new target motion. For a query consisting of a source clip, a target skeleton, and an action request, ANCHOR first predicts a coarse source action cluster using a random-forest classifier [Breiman, 2001] trained on motion-space descriptors from the training skeletons. It then searches the target skeleton’s motion library and returns one existing clip.

Table 9: SIF on the main evaluation (1,891 triples). Raw and length-controlled SIF with 95% intervals that resample whole source skeletons; p is the one-sided shuffle-test p -value; variation is the median ratio of output spread to source spread under raw scoring; n is the number of triples a method supports. The last row is the ACE ablation of Appendix I, not an evaluated method.

Method	n	Raw SIF [95% CI]	p	Length-controlled SIF [95% CI]	p	Variation	Reading
ACE-I	1,891	+0.420 [+0.31, +0.53]	<0.001	+0.248 [+0.13, +0.38]	<0.001	0.03	above floor
MoReFlow-T	1,891	+0.255 [+0.19, +0.33]	<0.001	+0.246 [+0.17, +0.32]	<0.001	0.20	above floor
MoReFlow-I	1,891	+0.257 [+0.19, +0.33]	<0.001	+0.244 [+0.18, +0.32]	<0.001	0.16	above floor
ACE-T	1,891	+0.336 [+0.24, +0.44]	<0.001	+0.180 [+0.05, +0.31]	0.009	0.02	above floor
Motion2Motion-Direct	1,891	+0.143 [+0.08, +0.22]	<0.001	+0.129 [+0.06, +0.20]	<0.001	1.52	above floor
Motion2Motion-BVH	1,872	+0.032 [0.000, +0.07]	0.024	+0.032 [-0.003, +0.07]	0.016	0.79	near floor
random-same-exact-action	1,864	+0.002 [-0.01, +0.02]	0.413	+0.018 [-0.03, +0.07]	0.293	0.13	at floor
AL-Flow	1,683	+0.020 [-0.02, +0.06]	0.114	+0.017 [-0.02, +0.05]	0.135	0.03	at floor
random-same-cluster	1,883	-0.023 [-0.05, -0.002]	0.958	+0.001 [-0.03, +0.03]	0.475	0.44	at floor
DPG-SB-v3	1,891	+0.016 [-0.01, +0.04]	0.158	-0.008 [-0.04, +0.02]	0.700	0.22	at floor
ANCHOR	1,891	+0.006 [-0.03, +0.05]	0.387	-0.010 [-0.05, +0.03]	0.683	0.00	at floor
AL-Flow-Src	1,413	-0.012 [-0.04, +0.02]	0.744	-0.026 [-0.06, +0.01]	0.928	0.07	at floor
AL-Flow-Src-G	1,891	-0.024 [-0.06, +0.01]	0.931	-0.029 [-0.06, +0.004]	0.966	0.03	at floor
AnyTop	1,891	-0.048 [-0.12, +0.03]	0.915	-0.061 [-0.14, +0.01]	0.954	1.50	at floor
ACE trained with $z_{src} = 0$ (real z_{src} at generation)	1,891	+0.371 [+0.25, +0.49]	<0.001	+0.198 [+0.07, +0.34]	0.006	0.02	ablation

Table 10: SIF on the original evaluation (49 triples, one target per source group). Columns as in Table 9; here no source clip is shared between triples.

Method	n	Raw SIF	p	Length-controlled SIF	p	Variation
MoReFlow-T	49	+0.203	0.011	+0.220	0.005	0.25
ACE-T	49	+0.385	<0.001	+0.205	0.013	0.02
ACE-I	49	+0.484	<0.001	+0.178	0.022	0.02
MoReFlow-I	49	+0.137	0.061	+0.160	0.038	0.28
Motion2Motion-Direct	49	+0.053	0.290	+0.091	0.146	1.69
ANCHOR	49	+0.014	0.380	+0.035	0.226	0.00
random-same-exact-action	48	+0.096	0.049	+0.032	0.338	0.00
random-same-cluster	48	+0.073	0.170	+0.008	0.464	0.25
AL-Flow-Src	38	+0.070	0.258	-0.009	0.534	0.11
DPG-SB-v3	49	-0.058	0.756	-0.031	0.644	0.22
AL-Flow	45	-0.066	0.769	-0.044	0.682	0.04
AnyTop	49	-0.033	0.639	-0.065	0.773	1.34
AL-Flow-Src-G	49	-0.028	0.629	-0.104	0.882	0.08
Motion2Motion-BVH	49	-0.085	0.823	-0.171	0.982	0.93

The motion descriptor, denoted $Q(x)$, is computed from joint-position trajectories rather than from a learned latent representation. It summarizes center-of-mass displacement and speed, heading velocity, contact timing, cadence, dominant limb usage, and global clip statistics. The candidate pool is the union of all target-skeleton clips in the predicted cluster and the ten clips with largest Q -descriptor similarity to the source. Candidate clips are scored by cluster agreement, descriptor similarity, and exact-action agreement, using the fixed comparator weights $(w_{\text{cluster}}, w_Q, w_{\text{action}}) = (1, 2, 3)$ recorded in the method roster.

This construction is why ANCHOR is gauge-invariant in the sense of Corollary 1.1: changing a learned latent parametrization cannot change a score computed directly from center-of-mass, heading, contact, cadence, and limb-usage measurements on the joint trajectories. At the same time, because ANCHOR returns an existing library clip, it does not claim to identify the source-conditioned retargeting map. Its purpose is to mark a diagnostic boundary, showing that action recovery can remain high at the source-blind floor. Table 19 records the action-cluster predictor, and Figure 15 shows how ANCHOR matches vary across action clusters.

H Action-Level AUC and Label Recovery

The standard cross-skeleton contrastive retrieval AUC is an action-level ranking test: it asks whether a generated target motion ranks positives above negatives in a target library. Section 5 shows that this test can accept label-only behavior on the covered queries. The appendix gives the broader enumeration and comparison tables behind that statement. Table 20 and Figure 16 report the full enumeration, while Table 21, Table 22, and Figure 17 collect the AUC-band controls.

Table 11: Clustered bootstrap intervals for SIF. Clustering by source skeleton, target skeleton, or action tests whether the positive conclusions are driven by a narrow subset of triples.

Method	each triple alone	by source skeleton	by target skeleton	by action
ACE-I	0.526 [0.351, 0.681]	0.526 [0.323, 0.709]	0.526 [0.365, 0.671]	0.526 [0.242, 0.629]
ACE-T	0.453 [0.293, 0.602]	0.453 [0.284, 0.618]	0.453 [0.274, 0.605]	0.453 [-0.009, 0.619]
MoReFlow-T	0.192 [-0.003, 0.380]	0.192 [-0.023, 0.386]	0.192 [-0.031, 0.405]	0.192 [-0.039, 0.357]
random-same-exact-action	0.185 [0.046, 0.326]	0.185 [0.069, 0.304]	0.185 [0.027, 0.354]	0.185 [0.000, 0.272]
Motion2Motion-Direct	0.131 [-0.061, 0.317]	0.131 [-0.098, 0.385]	0.131 [-0.075, 0.307]	0.131 [-0.720, 0.265]
random-same-cluster	0.104 [-0.095, 0.298]	0.104 [-0.057, 0.279]	0.104 [-0.115, 0.315]	0.104 [-0.272, 0.189]
AL-Flow-Src	0.076 [-0.108, 0.264]	0.076 [-0.109, 0.269]	0.076 [-0.112, 0.248]	0.076 [-0.222, 0.169]
MoReFlow-I	0.076 [-0.132, 0.279]	0.076 [-0.132, 0.271]	0.076 [-0.167, 0.319]	0.076 [-0.108, 0.474]
ANCHOR	0.040 [-0.068, 0.157]	0.040 [-0.078, 0.176]	0.040 [-0.068, 0.167]	0.040 [0.000, 0.046]
AnyTop	-0.010 [-0.185, 0.166]	-0.010 [-0.191, 0.166]	-0.010 [-0.207, 0.185]	-0.010 [-0.253, 0.142]
DPG-SB-v3	-0.019 [-0.215, 0.178]	-0.019 [-0.208, 0.188]	-0.019 [-0.244, 0.228]	-0.019 [-0.487, 0.045]
AL-Flow	-0.050 [-0.231, 0.137]	-0.050 [-0.187, 0.106]	-0.050 [-0.259, 0.161]	-0.050 [-0.239, 0.183]
AL-Flow-Src-G	-0.057 [-0.239, 0.130]	-0.057 [-0.207, 0.087]	-0.057 [-0.261, 0.128]	-0.057 [-0.523, 0.022]
Motion2Motion-BVH	-0.067 [-0.237, 0.110]	-0.067 [-0.247, 0.116]	-0.067 [-0.221, 0.089]	-0.067 [-0.320, 0.142]

Bold lower CI excludes zero. Bootstrap $N = 10,000$, seed = 42. ACE-I is the only method whose lower CI excludes zero under all four resampling schemes; ACE-T and random-same-exact-action survive resampling by source skeleton and by target skeleton, but resampling by action drops their lower CI to or below zero because the 37-triple intersection is concentrated on a few actions.

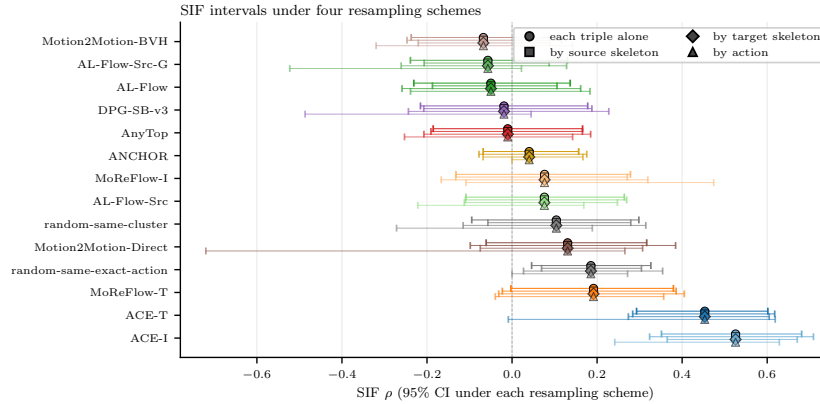


Figure 10: Clustered confidence intervals for SIF. The figure visualizes the dependence-aware intervals reported in Table 11.

I ACE Controls

ACE is the evaluated method with the largest raw SIF, so we traced where its signal comes from. This appendix removes or shuffles its source input, removes its adversarial loss in a matched comparison, and checks how its output variation should be measured.

ACE receives the source clip through a source latent, z_{src} , and in our evaluation each output also took its length from its source clip. Table 23 varies both inputs at generation on the 49 original triples, for ACE-T and for an ablation trained with z_{src} set to zero. That ablation still received the real source latent at generation and copied the source length, so its raw SIF of about +0.42 comes from these two inputs. When the source latent is removed at generation and the length is fixed, both models score exactly zero and their outputs no longer vary. This is the reference point for the source-blind floor. With the source removed but the length copied, both still score about +0.30, the share carried by length alone. Shuffling the source among the clips of a triple brings the score to zero or below. With the real source and a fixed length, ACE-T keeps +0.29, which is +0.29 above the removed-source result (95% CI [+0.12, +0.47]). Three training seeds give $+0.29 \pm 0.07$.

To test whether the adversarial loss creates this signal, we compare ACE-I with a model trained without that loss but otherwise identically: the same 60 training skeletons, the same data, and the same three seeds. Table 24 reports both scorings. When the output length follows the source, the model with the adversarial loss scores higher. When the length is fixed at generation, the difference

Table 12: Paired SIF differences on identical triples, with 95% intervals that resample whole source skeletons and two-sided sign-flip p -values that flip all triples of a source skeleton together.

Evaluation	Scoring	ACE-T minus MoReFlow-T	ACE-I minus MoReFlow-I
1,891 triples	raw	+0.081 [−0.02, +0.19], $p=0.139$	+0.163 [+0.05, +0.28], $p=0.008$
1,891 triples	length-controlled	−0.066 [−0.19, +0.06], $p=0.325$	+0.004 [−0.10, +0.11], $p=0.935$
49 triples	raw	+0.182 [−0.06, +0.44], $p=0.156$	+0.347 [+0.11, +0.60], $p=0.009$
49 triples	length-controlled	−0.015 [−0.21, +0.19], $p=0.888$	+0.018 [−0.21, +0.24], $p=0.880$

Table 13: Noise of a single triple and stability of the average. The first column is the standard deviation of SIF across three-clip subsets of five-clip triples; the other columns are standard deviations of the average over k randomly drawn triples.

Method	One triple (three clips)	Average over k triples			
		$k = 25$	$k = 100$	$k = 500$	$k = 1,000$
AnyTop	0.60	0.122	0.060	0.021	0.012
ACE-T	0.59	0.124	0.060	0.022	0.013
ACE-I	0.56	0.110	0.054	0.019	0.012
MoReFlow-T	0.57	0.123	0.059	0.023	0.013
random-same-exact-action	0.31	0.089	0.046	0.017	0.010

disappears on both evaluation sets. The difference therefore depends on the output length following the source. Without the adversarial loss, the outputs also vary more. A further ablation without the source-feature loss diverged during training and gave no usable model.

Output variation needs care under length control. Resampling finished outputs to 64 frames stretches clips of different lengths by different amounts, which can make identical motions look different. Table 25 shows that ACE’s outputs, nearly identical in raw scoring, gain a variation of 0.14 to 0.23 after stretching. Identical copies of a single ACE output, cut to the lengths of its siblings, reach almost the same value. MoReFlow’s variation is the same with and without stretching and clearly exceeds what stretching alone produces. We therefore report ACE’s variation in raw scoring and with the length fixed at generation.

The nearest-neighbor leakage check in Table 26 addresses a different explanation: that ACE could be lifting SIF by memorizing source clips. Most ACE outputs are closer to a target-pool training clip than to the corresponding source clip in the reported four-dimensional kinematic feature space, which weakens source memorization as the explanation for the SIF lift.

J Human-to-Robot Studies

Truebones has no true correspondences, so it can show that methods sit at the source-blind floor but not what a correct map would score. The two studies in this appendix supply true pairs. They test SIF where the answer is known, and they train the two objective classes of Section 4 on the same data as a model given the true pairs.

Data. The first study uses BONES-SEED [Bones Studio], a motion-capture dataset whose human clips come with retargeted counterparts on the Unitree G1 humanoid (Motion Data by Bones Studio, <https://bones.studio/>). We selected 90 action groups with three or four clips each, 355 clips in total. The second study starts from the 77 recordings of LAFAN1 [Harvey et al., 2020]. We cut each recording into 10-second clips and grouped the clips by routine; each routine is performed by up to five people, giving 29 groups. We converted every clip to six humanoid robots with GMR [Araujo et al., 2025], a public inverse-kinematics retargeting tool: Unitree G1, Booster T1, Fourier N1, Stanford Toddy, EngineAI PM01, and PAL Talos, which differ in height and have 23 to 31 joints. The converted clip serves as the true counterpart. LAFAN1 is released under CC BY-NC-ND 4.0; the paper shows rendered stills for illustration and does not release the retargeted clips. GMR is released under the MIT License, and the robot models are those distributed with GMR, credited to their original sources in its repository. We use two settings: three clips per group, as in Truebones (72 clips; 17 groups have at least three), and a denser one (154 clips; 26 groups). All six-robot outputs

Table 14: Generation noise control on the original 49 triples. Independent noise uses three seeds; shared noise reuses one noise sample within each triple.

Method	Independent noise, three seeds			Mean $\pm$ sd	Shared noise
	seed 1	seed 2	seed 3		
AnyTop	−0.033	+0.040	−0.098	−0.031 $\pm$ 0.069	+0.243
AL-Flow	−0.066	−0.043	+0.066	−0.015 $\pm$ 0.070	+0.002
AL-Flow-Src	+0.070	+0.049	−0.140	−0.007 $\pm$ 0.115	+0.199
AL-Flow-Src-G	−0.028	+0.048	+0.075	+0.031 $\pm$ 0.053	+0.239
DPG-SB-v3	−0.058	+0.138	+0.052	+0.044 $\pm$ 0.098	+0.064
MoReFlow-T	+0.203	+0.203	+0.203	+0.203 $\pm$ 0.000	deterministic
MoReFlow-I	+0.137	+0.137	+0.137	+0.137 $\pm$ 0.000	deterministic

Table 15: Pair counts per SIF triple. The minimum and median values are fixed by the eligibility rule; the few larger cells determine how much within-cell geometry is available.

Method	Min pairs	Median pairs	Max pairs	n triples
ACE-I	3	3	5	49
ACE-T	3	3	5	49
AL-Flow	3	3	5	45
AL-Flow-Src	3	3	5	38
AL-Flow-Src-G	3	3	5	49
ANCHOR	3	3	5	49
AnyTop	3	3	5	49
DPG-SB-v3	3	3	5	49
Motion2Motion-BVH	3	3	5	49
Motion2Motion-Direct	3	3	5	49
MoReFlow-I	3	3	5	49
MoReFlow-T	3	3	5	49
random-same-exact-action	3	3	5	48
random-same-cluster	3	3	5	48

Min and median are uniformly 3 because the SIF benchmark requires ≥ 3 source clips per triple by design; max reflects the few triples in Truebones with 4 or 5 source clips. Variation across methods comes from the few queries that some baselines fail to score.

are scored on a common 256-frame window. In both studies the sparsity is created by us: the pairs exist, but the two theorem objectives never see them.

Models. Figure 18 shows the data and the trained models on the robots. In both studies we train one small transformer (four layers, width 256) in three ways, with the same data, number of steps, and optimizer. The unpaired objective couples two variational autoencoders, one per body, through a shared Gaussian latent and trains each only on its own motions, which places it in the class of Theorem 1. The averaging objective regresses each human clip, with squared error, onto a clip of the same group on the robot, drawn at random at every step, which places it in the class of Proposition 1.1. The true-pair model uses the same regressor trained on the true counterpart. Two reference rows need no training: the true counterpart itself and a random clip from the same group.

Human-to-G1 results. Table 27 extends Table 2 with length-controlled SIF. The true counterpart scores the same with and without length control (0.960 and 0.959), so SIF on real pairs does not depend on clip length. When scoring action AUC, each output’s own clip is removed from the reference set, as the query’s clip is for every output.

Six-robot predictions. For the six-robot study we wrote down five predictions with numeric thresholds before scoring any trained model; we had seen the reference rows for one robot during a pilot run. Here Q is a model’s output variation divided by that of the true counterpart, matched group by group. Each prediction is checked 24 times: six robots, two settings, and raw or length-controlled scoring. Table 28 reports the outcome, including the misses, and Table 29 gives every robot. The

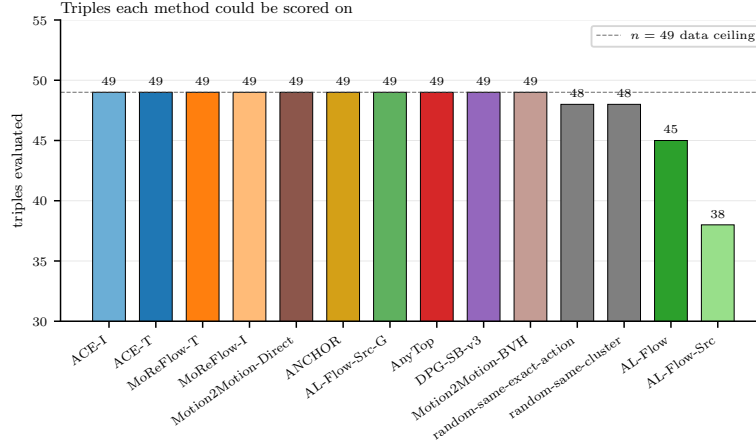


Figure 11: Pair-count distribution for SIF-eligible triples.

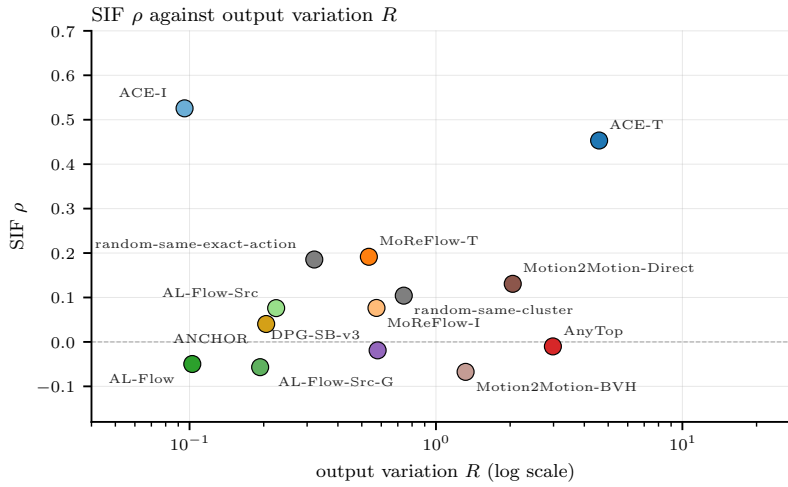


Figure 12: SIF versus output variation (the diversity ratio R). High output variation is not sufficient for source-instance preservation, because the geometry must align with the source-side geometry after the target skeleton and action are fixed.

true counterpart falls just short of the 0.80 threshold on Booster T1 (0.79) and in one Stanford Toddy evaluation (0.77), but it is clearly above zero everywhere. The random-clip prediction holds in every denser evaluation; with three clips per group its intervals are too wide to decide. The unpaired objective keeps about 40 percent of the variation on PAL Talos with three clips per group, and collapses in every other evaluation.

An adversarial objective without pairs. We also trained an objective in the style of ACE on the human-to-G1 data: a generator with a motion-space discriminator and a source-feature loss, without pairs. The planned run diverged: its outputs grew without bound, and none passed the realism check. We then re-ran it post hoc with ACE’s own optimizer settings, which stabilized training. Table 30 reports all three seeds. No seed scores well on correlation, action accuracy, variation, and realism together.

Four measures on identical test items. Table 31 reports SIF, action AUC, output variation, and realism together for the Truebones methods on the 49 original triples. No method is high on both SIF and action AUC: the retrieval and label-only rows reach the highest action AUC with SIF near zero, and ACE reaches the highest SIF with chance-level action AUC.

Table 16: SIF and L-SIF for each method. L-SIF is a latent-space diagnostic and is therefore reported only where the representation is comparable across queries.

Method	SIF ρ	L-SIF ρ
ACE trained with $z_{\text{src}} = 0$	0.423	0.037
ACE-I (3 seeds avg)	0.410	0.049
ACE-T (3 seeds avg)	0.408	0.088
ACE without adversarial loss (3 seeds avg)	0.214	0.119
MoReFlow-T	0.192	-0.200
MoReFlow-I	0.076	-0.205
AL-Flow-Src	0.076	-0.113
AnyTop	-0.010	0.451
AL-Flow	-0.050	-0.098
AL-Flow-Src-G	-0.057	-0.126

Cross-method Pearson correlation between SIF and L-SIF: $r = 0.199$, $p = 0.46$ across $n = 16$ method runs. The 95% interval of L-SIF lies above zero for AnyTop and below zero for MoReFlow-I and MoReFlow-T; it includes zero for every other run.

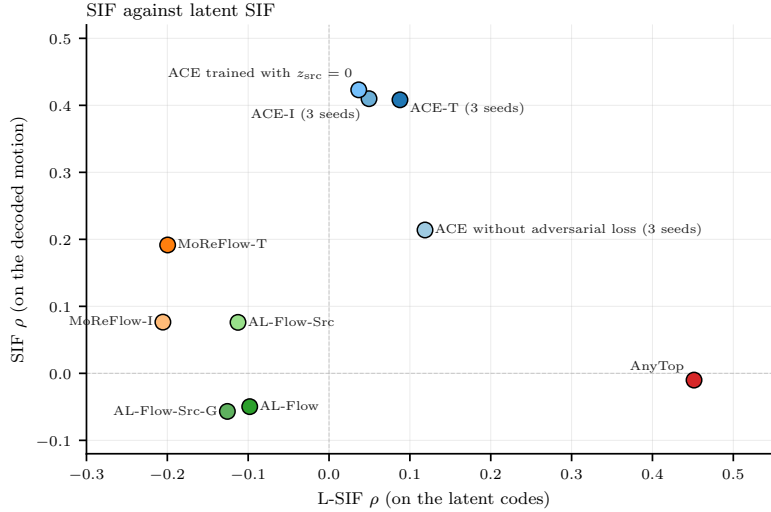


Figure 13: SIF and L-SIF decoupling. Source-instance fidelity in decoded motion need not coincide with a positive latent-space correlation.

Table 32 gives the same measures for the six-robot study. The model trained on true pairs matches the true counterpart on all four. The averaging objective keeps little of the variation and produces many unrealistic poses, and the unpaired objective falls to chance on action AUC. Action AUC is lower for every case here, including the true counterpart, because several LAFAN1 categories are all forms of locomotion. Action AUC and realism for this study were computed after the pre-registration, with the same definitions as the human-to-G1 study.

K Qualitative Evidence and Supplementary Videos

The qualitative figures are not used as quantitative proof, but they make the SIF failure modes visible. The appendix provides three full-grid still figures that correspond to the supplementary videos. Each grid fixes a source-target triple and compares the evaluated methods across three source clips, so rows that move in near lockstep indicate source-blind behavior while rows with source-dependent variation indicate partial source conditioning. Figure 4 gives the compact method panorama in the main text. Figures 19, 20, 21, and 22 provide the appendix still-image views corresponding to this qualitative protocol.

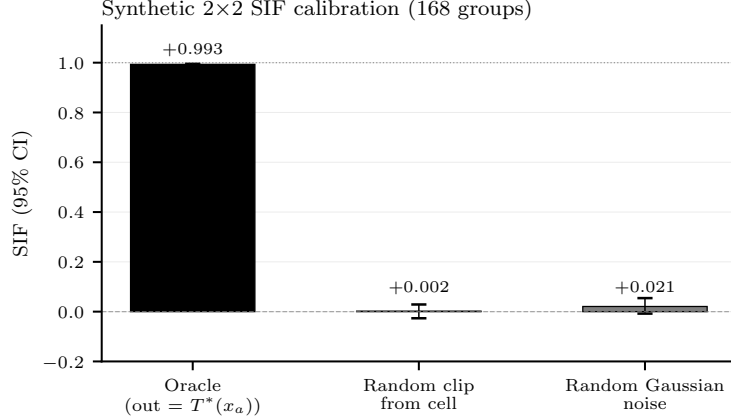


Figure 14: Synthetic 2×2 SIF calibration. The oracle uses the known source-conditioned transport $T^*(x_a)$; the random-clip and Gaussian references deliberately ignore the source instance.

Table 17: Synthetic SIF calibration. The oracle lies near perfect source-instance preservation, while source-blind references remain near zero.

Output	SIF (95% CI)	p	Variation
Oracle (output = $T^*(x_a)$)	+0.993 [+0.992 , +0.995]	<0.001	0.94
Random clip from cell	+0.002 [−0.027, +0.029]	0.456	0.96
Random Gaussian noise	+0.021 [−0.008, +0.054]	0.164	1.3×10^5

Synthetic 2×2 paired-dense setting, 168 groups of three to six source clips. Another 168 groups are left out because their source clips are identical once rotation and scale are removed, so SIF is undefined there. The oracle saturates SIF; both source-blind references sit at zero. Noise has no motion structure, so its variation is very large.

The supplementary material contains nine companion videos: two main-body videos for the panorama and failure-mode figures, and seven appendix videos for additional source-target triples. The videos use the same projection rule as the still figures and synchronize playback phase across methods, making source-blind lockstep behavior and source-conditioned variation easier to see.

L Latent Schrödinger Bridge Comparator Scope

The evaluated latent Schrödinger bridge comparator, reported under the public method name DPG-SB-v3 in the roster, uses retrieval-initialized noise, a shared conditioned bridge, a target-side latent constraint, and a latent-space cycle term. The decoded-motion bone, contact, and smoothness penalties were not included in this evaluation. This caveat is important for interpretation: DPG-SB-v3 tests where the evaluated latent bridge variant lies relative to the source-blind floor, not an exhaustive set of motion-space penalties for bridge objectives.

M Potential Societal Impacts

Cross-skeleton motion generation may broaden access to motion content in animation, education, assistive design, and scientific visualization by reducing the need to record every action separately for every body. In these settings, a source-preserving model could help creators transfer motion intention across characters, support low-resource motion libraries, and make comparative studies of movement easier. This paper contributes a diagnostic and theoretical account of when such transfer claims are not identified by standard evidence, rather than a deployed generator.

The same capability also carries risks. More reliable motion transfer could make synthetic embodied media easier to produce, including misleading performances attributed to people or characters who

Table 18: Conditional-mean degeneration in the controlled 2×2 setting. The predicted variance remains a small fraction of oracle variance even when the number of random target-cell pairs increases.

M pairs / cell	$\sigma_{\text{pred}}^2 / \sigma_{\text{oracle}}^2$	MSE / σ_{oracle}^2	n_{seeds}
2	0.020 ± 0.006	1.45 ± 0.08	3
4	0.014 ± 0.002	1.29 ± 0.03	3
8	0.020 ± 0.006	1.34 ± 0.08	3
16	0.015 ± 0.002	1.31 ± 0.12	3
32	0.017 ± 0.001	1.22 ± 0.05	3
50	0.016 ± 0.003	1.26 ± 0.09	3

Each source clip is trained against the true target of another clip from the same cell, drawn afresh at every step. Across all M and 3 seeds, the predicted variance stays at about 2% of the oracle within-cell variance, the cell-mean compression Proposition 1.1 predicts.

Table 19: ANCHOR action-cluster predictor. The held-out row shows that the cluster prediction itself is imperfect, which is why ANCHOR is interpreted as a comparator rather than an oracle.

Domain	Cluster classifier accuracy	n correct / total
In-domain (train skeletons)	99.8%	543 / 544
Held-out (reserved cross-skeleton)	61.5%	59 / 96

Replacing the ground-truth cluster label with the predicted cluster pulls Procrustes performance below random on exact-tier and on a hand-curated query subset. The in-domain row is the accuracy on the clips the classifier was trained on.

did not enact them, and action-level evaluation may overstate reliability when the source-conditioned map is not actually preserved. Biases in the source and target motion libraries may also be propagated across bodies, especially when some anatomies, actions, or movement styles are sparsely represented. Our analysis supports a conservative evaluation stance: cross-skeleton methods should report source-instance diagnostics alongside action-level scores, disclose the motion domains and licenses on which they are trained, and avoid claims of retargeting fidelity when only target-action recovery has been established.

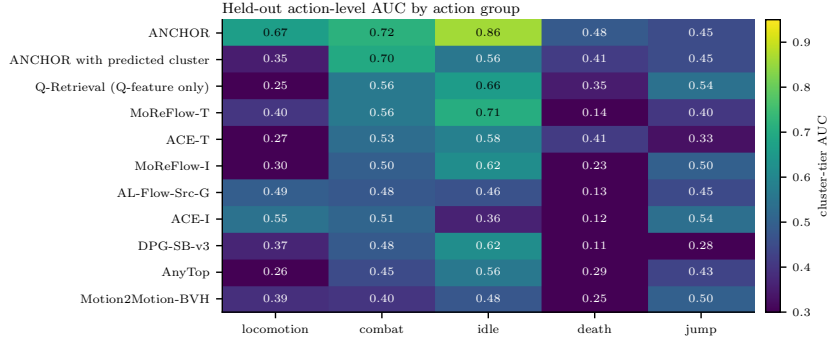


Figure 15: Per-cluster ANCHOR matches. The comparator’s behavior varies across action clusters, reinforcing that it characterizes a floor rather than a universal retargeting rule.

Table 20: Full ANCHOR enumeration across cluster-eligible triples. The held-out subsets show that strong action-level ranking can arise from target-library label structure without identifying a source-conditioned map.

Subset	n triples (cluster-eligible)	Cluster-tier AUC (95% CI)
All	30,029	0.824 [0.820, 0.829]
In-distribution (same train pair)	21,838	0.845 [0.841, 0.849]
Mixed (one held-out skeleton)	7,622	0.774 [0.762, 0.787]
Held-out (both cross-skeleton)	569	0.711 [0.669, 0.758]

ANCHOR on the 30,497-pair full enumeration. Cluster-eligibility filter removes triples whose target skeleton has no in-cluster positives. The mixed subset is the held-out case.

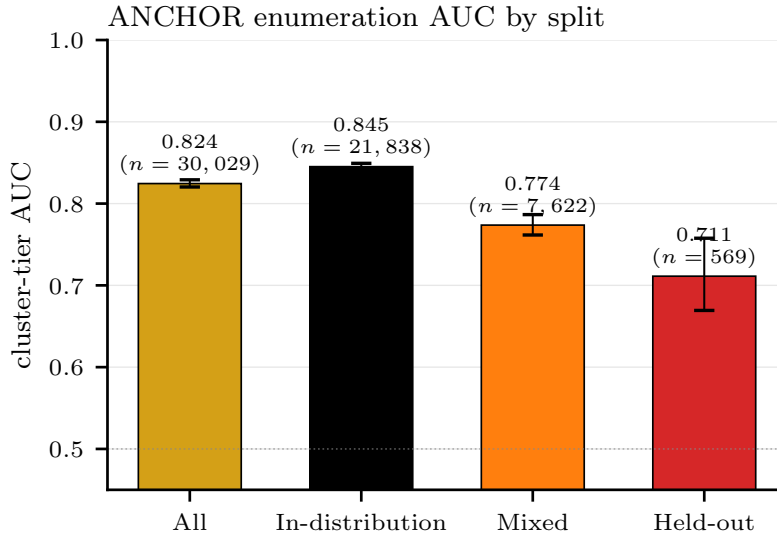


Figure 16: Enumeration breakdown for ANCHOR. The large enumeration separates in-distribution, mixed, and held-out skeleton regimes.

Table 21: Action-level AUC for generated variants and supervision-matched comparators. These values are not interpreted as source-preserving transfer; they show how much action-level evidence can be explained without SIF.

Variant	Conditioning channels	Cluster AUC
AnyTop	self-supervised source-conditioned diffusion	0.465
Precursor action-conditioned generator	action-conditioned generation	0.485
Precursor latent-bridge generator	latent bridge objective	0.483
DPG-SB-v3	latent bridge objective; no decoded-motion penalties	0.447
AL-Flow	cluster + exact-action labels	0.538
AL-Flow-Src	AL-Flow + source motion + skeleton identity	0.542
AL-Flow-Src-G	AL-Flow + source motion + skeleton graph	0.470

The two precursor rows are reported only for the action-level AUC analysis and are not part of the 14-method SIF roster. AL-Flow and AL-Flow-Src produce outputs for 159 and 100 of the 300 queries in each fold, and their AUC is computed on those queries.

Table 22: Master action-level AUC comparison on held-out queries. The table reports cluster-tier AUC under three distances: Procrustes trajectory distance; Z-DTW, dynamic time warping on z-normalized body-part trajectories; and Q-comp, a combination of centre-of-mass path, foot-contact timing, cadence, and limb-usage differences. Library-based references are listed separately from generative methods.

Cluster-tier held-out AUC (average of folds 42 and 43)			
Method	Procrustes	Z-DTW	Q-comp
Action oracle	0.873	0.846	0.894
Self-positive reference	0.860	0.847	0.909
ANCHOR	0.681	0.704	0.728
Cluster-Classifer Retrieval	0.646	0.669	0.693
Q-Retrieval (Q-feature only)	0.453	0.523	0.597
MoReFlow-T	0.463	0.496	0.558
MoReFlow-I	0.419	0.431	0.567
ACE-T	0.448	0.451	0.529
ACE-I	0.401	0.469	0.550
Random target skeleton (null)	0.474	0.510	0.530

Motion2Motion-BVH has Procrustes only and is reported separately.

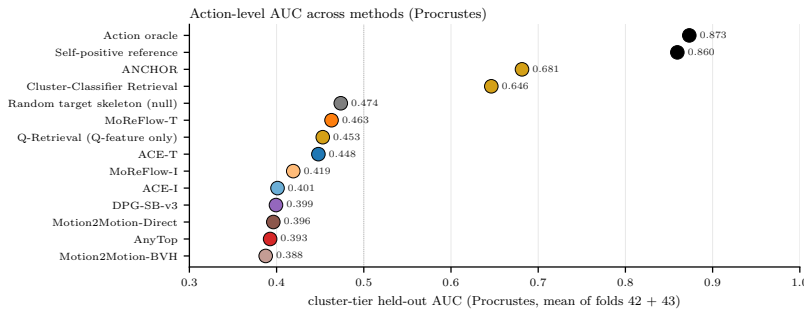


Figure 17: Action-level AUC band. Many methods occupy a narrow action-ranking band, which motivates SIF as the missing source-instance axis.

Table 23: Removing or shuffling ACE’s source input at generation (49 original triples). Removed means z_{src} set to zero at generation; shuffled means each output receives the source of another clip in the same triple; fixed means every output is generated with 64 frames. p is the shuffle-test p -value; variation as in Table 9.

Source at generation	Output length	ACE-T			Model trained with $z_{\text{src}} = 0$		
		SIF	p	Variation	SIF	p	Variation
real	copied	+0.385	<0.001	0.020	+0.419	<0.001	0.023
real	fixed	+0.294	<0.001	0.003	+0.213	0.006	0.004
removed	copied	+0.300	<0.001	0.009	+0.299	<0.001	0.007
removed	fixed	0.000	1.000	0.000	0.000	1.000	0.000
shuffled	copied	−0.192	0.987	0.020	−0.165	0.970	0.023
shuffled	fixed	−0.064	0.774	0.003	+0.040	0.328	0.004

Table 24: ACE-I with and without the adversarial loss, trained identically otherwise (three seeds each). SIF is averaged over seeds; variation gives the range of the per-seed medians; the difference is paired over triples, with a 95% interval that resamples whole source skeletons.

Scoring	Triples	With adversarial loss		Without		Difference [95% CI]	p
		SIF	Variation	SIF	Variation		
raw, length copied	49	+0.407	0.017–0.022	+0.228	0.271–0.906	+0.179 [+0.04, +0.33]	0.022
length fixed at generation	49	+0.292	0.002–0.004	+0.266	0.073–0.107	+0.026 [−0.09, +0.15]	0.677
length fixed at generation	1,891	+0.329	0.002–0.003	+0.329	0.078–0.111	0.000 [−0.05, +0.06]	0.999

Table 25: Output variation after stretching every output to 64 frames (49 original triples). Identical copies are one output cut to the lengths of its siblings, so any variation they show comes from stretching alone.

Method	Actual outputs		Identical copies, stretched to 64 frames
	Raw	Stretched to 64 frames	
ACE-T	0.020	0.231	0.149
ACE-I	0.020	0.135	0.142
MoReFlow-T	0.247	0.217	0.093
MoReFlow-I	0.276	0.279	0.078

Table 26: ACE nearest-neighbor leakage check. Distances are computed in a kinematic feature space; the majority of ACE outputs are closer to the target pool than to the source clip.

Method	Queries	Closer to the target pool	Mean distance to source	Mean distance to target pool	Ratio
ACE-I	130	103 (79.2%)	3.449	2.215	1.56
ACE-T	130	99 (76.2%)	3.697	2.650	1.39

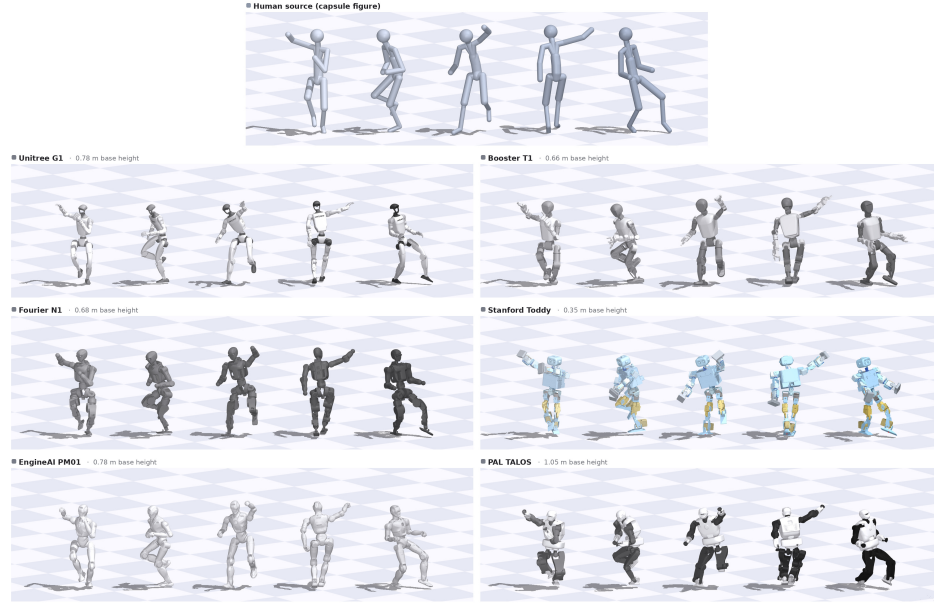
Distance computed in a four-dimensional kinematic feature space (centre-of-mass displacement, centre-of-mass variance, mean velocity, foot-contact density). 76–79% of ACE outputs are closer to a target-pool training clip than to the corresponding source clip, weakening source-memorisation as an explanation of ACE’s SIF lift.

Table 27: Human-to-G1 study, all four measures on identical test items (90 groups, 355 clips), with SIF under both scorings. Variation is the median ratio of output spread to source spread; realistic as in Table 2.

	Raw SIF	Length-controlled SIF	Action AUC	Variation	Realistic
True retargeted counterpart	+0.960	+0.959	0.980	0.518	97.2%
Random same-action clip	−0.008	−0.068	0.978	0.414	97.5%
Unpaired objective	+0.203	+0.154	0.499	0.001	100.0%
Averaging objective	+0.379	+0.343	0.820	0.048	53.0%
Model trained on true pairs	+0.900	+0.698	0.967	0.521	94.6%

A One dance, six robots

LAFAN1 dance routine, 10 s; five key poses, earliest faded; true retarget on every robot



B Two performers, same routine, on Unitree G1

same routine and 256-frame window for both performers; same five key poses and camera in every row

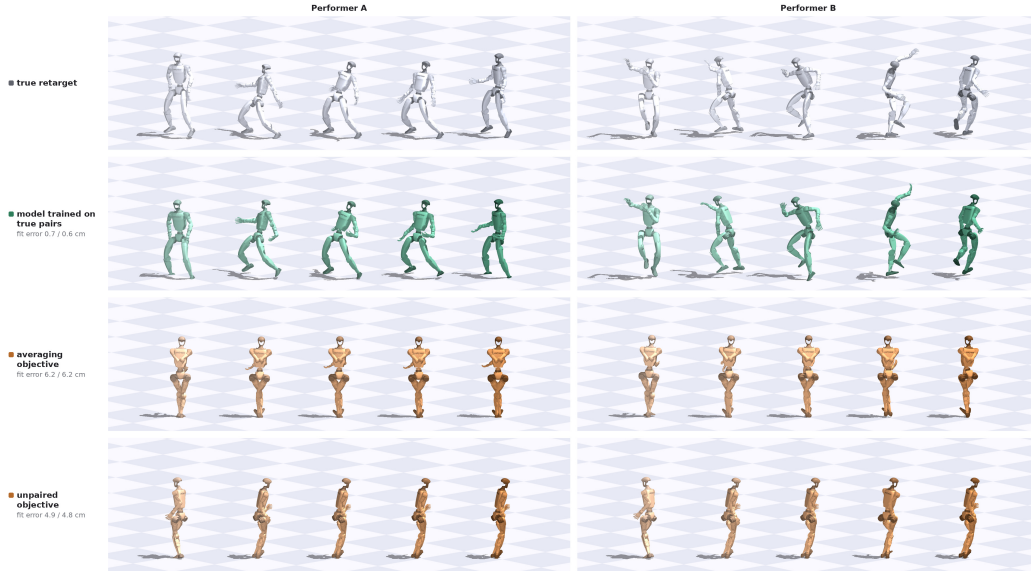


Figure 18: Human-to-robot renders. (A) One LAFAN1 dance routine (10 s) retargeted by GMR to six humanoid robots; five key poses per strip, earliest faded. Each strip is framed to its robot, so compare poses, not sizes; the base heights give the true scale. (B) Two performers dancing the same routine, and what each model produces for each of them on the Unitree G1. The true retargets and the model trained on true pairs differ between the performers. The averaging and unpaired objectives produce nearly the same motion for both, and that motion barely changes over time. Model outputs are predicted body positions, drawn as the nearest G1 pose; the fit error is listed under each row. The 5 to 6 cm errors of the two collapsed rows mean their raw predictions are less robot-like than drawn. The two performers are not aligned in time, so part of the difference in the top rows is timing rather than style.

Table 28: Six-robot predictions, fixed before any trained model was scored, and their outcomes over 24 evaluations (six robots, two settings, two scorings). Q is a model’s output variation divided by that of the true counterpart, matched group by group; intervals are group bootstraps.

Prediction	Threshold	Result
The true retarget is recognized	$\text{SIF} \geq 0.80$, interval above 0	19 of 24; every interval above 0; misses 0.768–0.799
A random same-group clip sits at the floor	90% interval within ± 0.20	12 of 24; 12 of 12 in the denser setting
The averaging objective loses variation	variation $Q \leq 0.35$, upper bound < 0.50	24 of 24
The unpaired objective loses variation	variation $Q \leq 0.35$, upper bound < 0.50	22 of 24; missed with variation $Q = 0.41$
The true-pair model recovers both	$\text{SIF} \geq 0.60$ and variation $Q \geq 0.70$	24 of 24; SIF 0.77–0.99

Table 29: Six-robot results for every robot, setting, and scoring. SIF columns give the correlation; Q columns give output variation relative to the true counterpart.

Robot	Setting	Scoring	True retarget	Random clip	Unpaired	Averaging	True-pair model	
			SIF	SIF	Variation (Q)	Variation (Q)	SIF	Variation (Q)
Unitree G1	three-clip	raw	+0.988	+0.066	0.059	0.286	+0.987	1.00
Unitree G1	three-clip	length-controlled	+0.987	+0.066	0.059	0.286	+0.987	1.00
Unitree G1	denser	raw	+0.966	−0.061	0.012	0.035	+0.965	0.99
Unitree G1	denser	length-controlled	+0.966	−0.058	0.012	0.035	+0.965	0.99
Booster T1	three-clip	raw	+0.799	+0.024	0.055	0.347	+0.845	0.99
Booster T1	three-clip	length-controlled	+0.788	+0.026	0.057	0.346	+0.842	0.99
Booster T1	denser	raw	+0.790	−0.062	0.036	0.067	+0.772	0.99
Booster T1	denser	length-controlled	+0.790	−0.059	0.036	0.067	+0.772	0.99
Fourier N1	three-clip	raw	+0.883	+0.098	0.027	0.278	+0.882	1.00
Fourier N1	three-clip	length-controlled	+0.883	+0.099	0.026	0.280	+0.881	1.01
Fourier N1	denser	raw	+0.899	−0.047	0.019	0.044	+0.895	1.00
Fourier N1	denser	length-controlled	+0.898	−0.044	0.019	0.044	+0.894	1.00
Stanford Toddy	three-clip	raw	+0.768	+0.182	0.017	0.267	+0.866	1.00
Stanford Toddy	three-clip	length-controlled	+0.877	+0.185	0.017	0.267	+0.872	1.00
Stanford Toddy	denser	raw	+0.861	−0.050	0.190	0.049	+0.862	0.99
Stanford Toddy	denser	length-controlled	+0.860	−0.047	0.191	0.049	+0.861	0.99
EngineAI PM01	three-clip	raw	+0.887	+0.099	0.316	0.294	+0.885	1.00
EngineAI PM01	three-clip	length-controlled	+0.885	+0.099	0.317	0.295	+0.883	1.00
EngineAI PM01	denser	raw	+0.906	−0.051	0.027	0.051	+0.907	0.99
EngineAI PM01	denser	length-controlled	+0.907	−0.048	0.027	0.051	+0.908	0.99
PAL Talos	three-clip	raw	+0.895	+0.027	0.405	0.260	+0.892	1.00
PAL Talos	three-clip	length-controlled	+0.896	+0.026	0.405	0.261	+0.892	1.00
PAL Talos	denser	raw	+0.878	−0.043	0.134	0.051	+0.878	0.99
PAL Talos	denser	length-controlled	+0.878	−0.041	0.134	0.051	+0.878	0.99

Table 30: An adversarial objective in the style of ACE on the human-to-G1 data, without pairs. The planned run diverged; the re-runs use ACE’s optimizer settings and are post hoc.

	Raw SIF	Action AUC	Variation	Realistic
Planned run	+0.209	0.500	diverged	0.0%
Re-run, seed 1	+0.696	0.850	0.561	14.4%
Re-run, seed 2	+0.528	0.551	0.093	78.0%
Re-run, seed 3	+0.574	0.513	0.101	0.0%

Table 31: Truebones methods on the 49 original triples: SIF (raw), action AUC, output variation, and realism on identical test items. Realism is undefined for Motion2Motion-BVH, whose outputs use a different joint layout.

Method	Raw SIF	Action AUC	Variation	Realistic
ACE-I	+0.484	0.539	0.02	91.8%
ACE-T	+0.385	0.538	0.02	92.4%
MoReFlow-T	+0.203	0.596	0.25	99.4%
MoReFlow-I	+0.137	0.578	0.28	99.4%
random-same-exact-action	+0.096	0.799	0.00	100.0%
random-same-cluster	+0.073	0.813	0.25	100.0%
AL-Flow-Src	+0.070	0.633	0.11	100.0%
Motion2Motion-Direct	+0.053	0.514	1.69	58.5%
ANCHOR	+0.014	0.828	0.00	100.0%
AL-Flow-Src-G	−0.028	0.606	0.08	100.0%
AnyTop	−0.033	0.518	1.34	52.0%
DPG-SB-v3	−0.058	0.684	0.22	100.0%
AL-Flow	−0.066	0.643	0.04	100.0%
Motion2Motion-BVH	−0.085	0.449	0.93	—

Table 32: Six-robot study, four measures on identical test items: mean over the six robots, with the range in parentheses. Variation is Q , a model’s output variation divided by that of the true counterpart. Action AUC and realism were computed after the pre-registration.

	Raw SIF	Action AUC	Variation (Q)	Realistic
<i>Three clips per group</i>				
True retargeted counterpart	+0.87 (+0.77+0.99)	0.61 (0.60–0.63)	1.00 (1.00–1.00)	99 (99–99)%
Random same-group clip	+0.08 (+0.02+0.18)	0.62 (0.61–0.63)	0.73 (0.69–0.77)	99 (99–100)%
Unpaired objective	+0.24 (+0.11+0.36)	0.50 (0.49–0.52)	0.15 (0.02–0.41)	98 (89–100)%
Averaging objective	+0.11 (+0.07+0.20)	0.60 (0.58–0.63)	0.29 (0.26–0.35)	46 (33–83)%
Model trained on true pairs	+0.89 (+0.85+0.99)	0.61 (0.60–0.63)	1.00 (0.99–1.00)	98 (97–99)%
<i>Denser setting</i>				
True retargeted counterpart	+0.88 (+0.79+0.97)	0.60 (0.58–0.61)	1.00 (1.00–1.00)	99 (99–99)%
Random same-group clip	−0.05 (−0.06+0.04)	0.59 (0.57–0.61)	0.82 (0.79–0.85)	98 (98–99)%
Unpaired objective	+0.15 (+0.10+0.20)	0.51 (0.50–0.52)	0.07 (0.01–0.19)	61 (1–100)%
Averaging objective	+0.16 (+0.12+0.18)	0.56 (0.55–0.59)	0.05 (0.03–0.07)	23 (8–84)%
Model trained on true pairs	+0.88 (+0.77+0.97)	0.60 (0.58–0.61)	0.99 (0.99–1.00)	99 (99–99)%

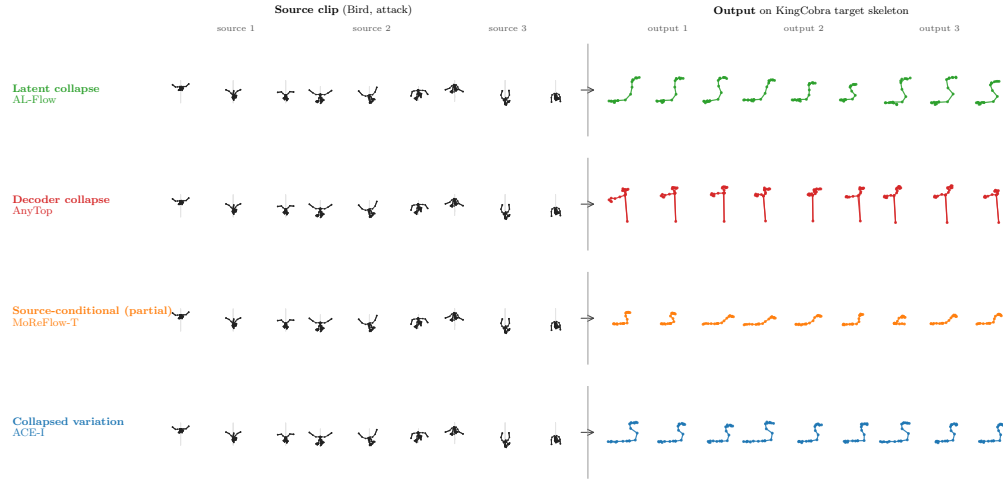


Figure 19: A source-clip substitution view on one Bird → KingCobra attack triple. The first three columns, labeled source 1 to source 3, are different Bird attack clips, while the target skeleton and action are fixed; the three columns on the right, labeled output 1 to output 3, show the corresponding KingCobra motions generated by each row. Each row is labeled by its source-instance failure mode: latent collapse for AL-Flow, where source identity is lost before decoding; decoder collapse for AnyTop, where latent variation is not expressed in the output; partial source-conditionality for MoReFlow-T, the clearest output-level source dependence among the displayed rows; and collapsed variation for ACE-I, whose outputs keep a positive correlation with their sources but remain nearly identical.



Figure 20: Full qualitative grid A. Three source clips are compared against target outputs from the evaluated methods on a fixed source-target-action triple.



Figure 21: Full qualitative grid B. The same layout is used to reveal whether output variation follows the source clips after target skeleton and action are fixed.

Hippopotamus → Lion, attack



Figure 22: Full qualitative grid C. The grid complements Figure 19 by showing all evaluated methods on an additional held-out triple.

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